BDA Course 2026

Lecture 2

Outline

- The Binomial model is very simple
  - used on its own:
    - coin tossing
    - chips from bag
    - COVID tests and vaccines
  - Commonly used as a building block, e.g., classification/logistic regression.
  - Useful to introduce observation model, likelihood, posterior, prior, marginalization, posterior summaries

Outline of Chapter 2

- 2.1 Binomial model (repeated experiment with binary outcome)
- 2.2 Posterior as compromise between data and prior information
- 2.3 Posterior summaries
- 2.4 Informative prior distributions (skip exponential families and sufficient statistics)
- 2.5 Gaussian model with known variance

Outline of Chapter 2 (cont)

- 2.6 Other single parameter models
  - the normal distribution with known mean but unknown variance is the most important
  - glance through Poisson and exponential
- 2.7 glance through this example, which illustrates benefits of prior information, no need to read all the details (it's quite long example)
- 2.8–2.9 Noninformative and weakly informative priors

Binomial: known θ

- Probability of event 1 in a trial: θ

Binomial: known θ

- Probability of event 1 in a trial: θ
- Probability of event 2 in a trial: $1 - \theta$

Binomial: known θ

- Probability of event 1 in a trial: θ
- Probability of event 2 in a trial: $1 - \theta$
- Probability of several events, assuming independence, is e.g. $\theta\theta(1-\theta)\theta(1-\theta)(1-\theta)...$

Binomial: known θ

- Probability of event 1 in a trial: θ
- Probability of event 2 in a trial: $1-\theta$
- Probability of several events, assuming independence, is e.g. $\theta\theta(1-\theta)\theta(1-\theta)(1-\theta)...$
- If there are n trials and order does not matter, then the probability that event 1 happens y times is

$$p(y|\theta,n) = \binom{n}{y}\theta^{y}(1-\theta)^{n-y}$$

Binomial: known θ

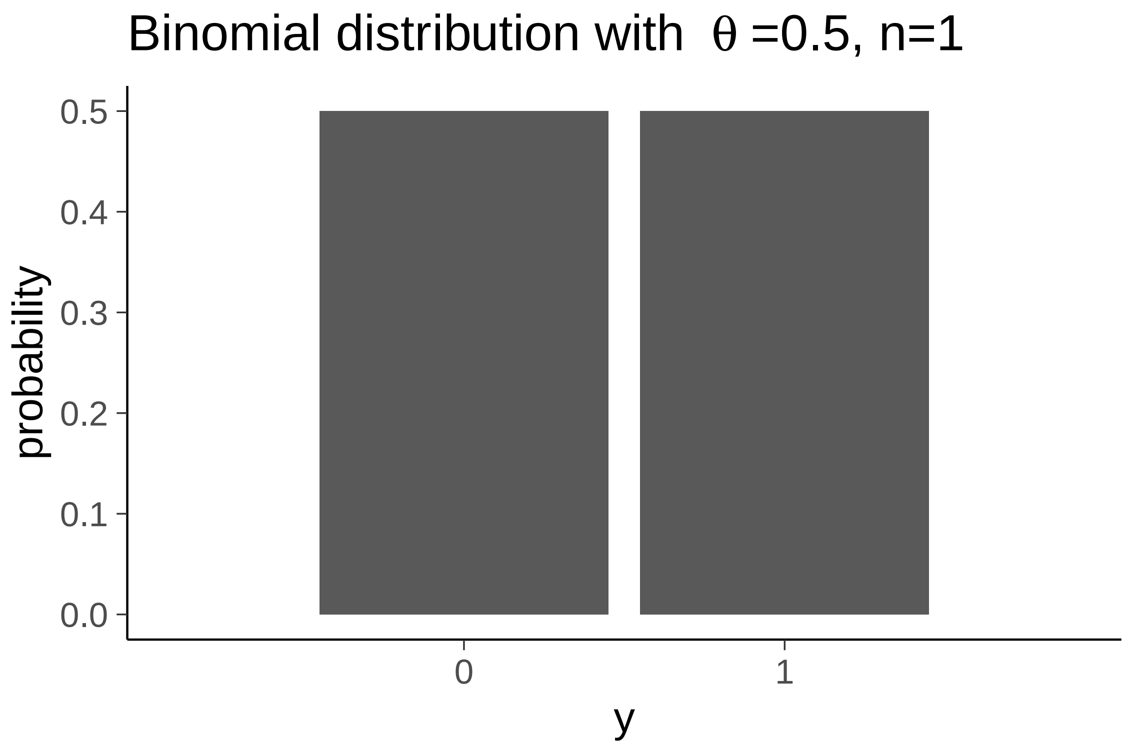
- Observation model (function of y, discrete)

$$p(y|\theta, n) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$

Binomial: known θ

- Observation model (function of y, discrete)

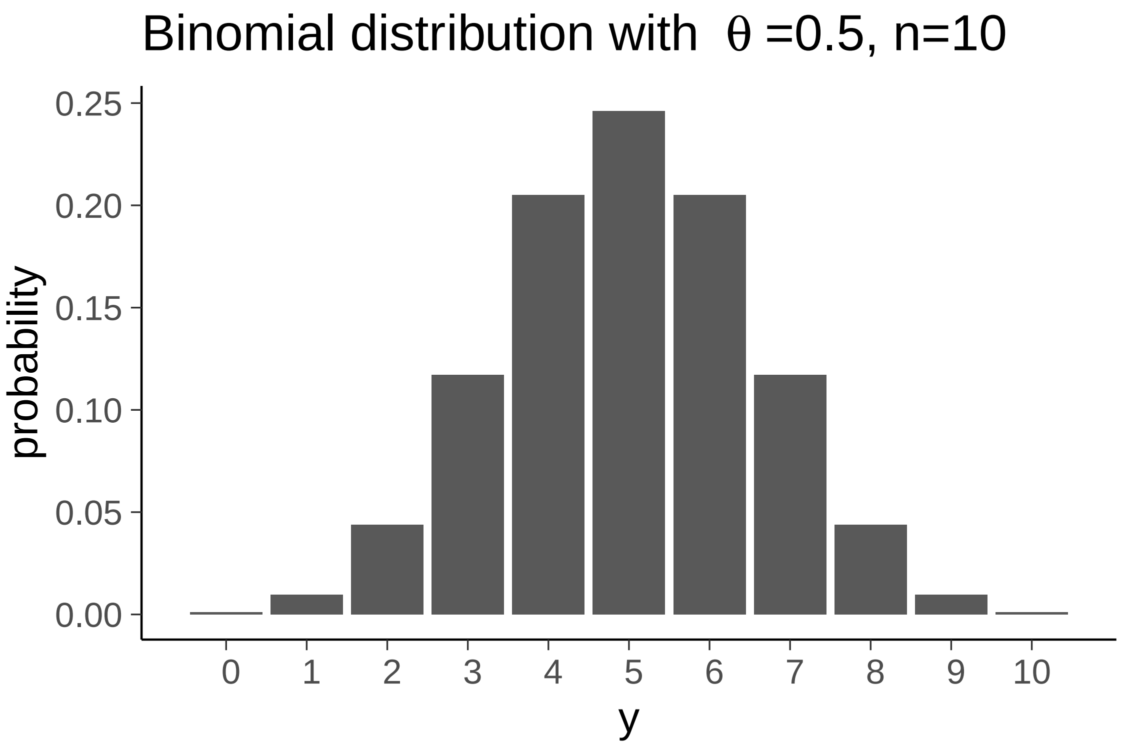
$$p(y|\theta, n) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$



Binomial: known θ

- Observation model (function of y, discrete)

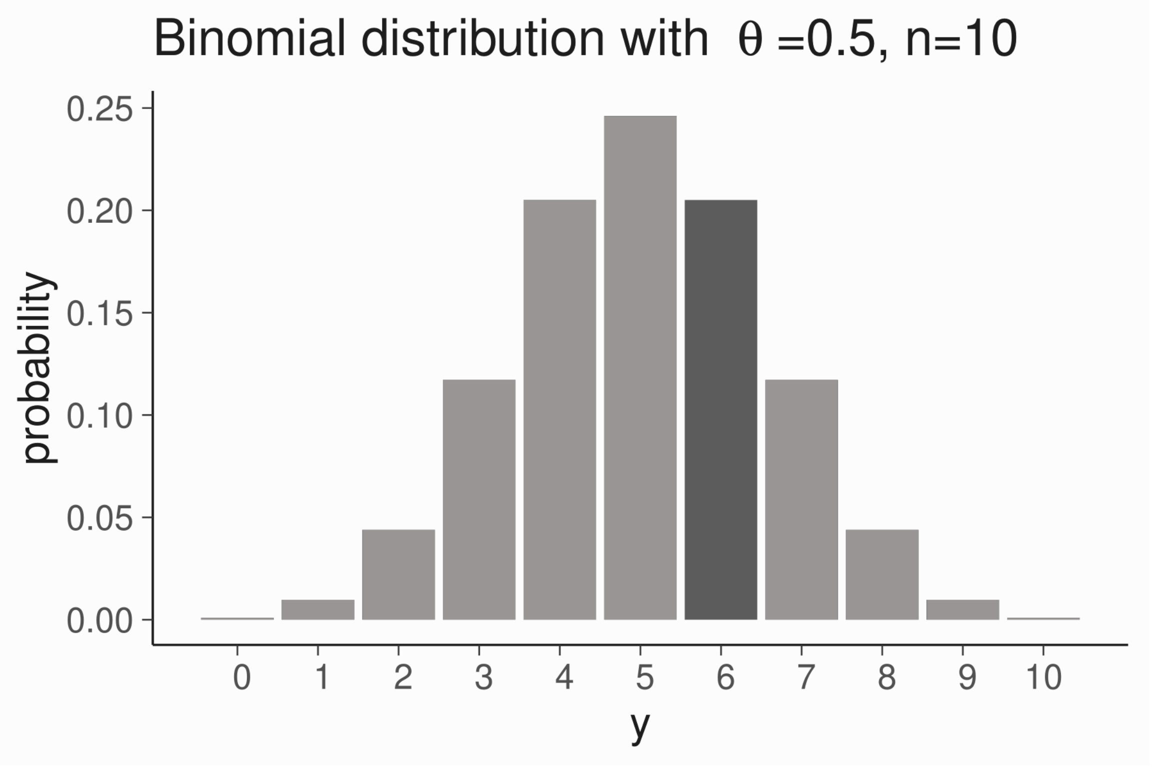
$$p(y|\theta, n) = \binom{n}{y} \theta^{y} (1-\theta)^{n-y}$$



Binomial, what if $y = 6$?

- Observation model (function of y, discrete)

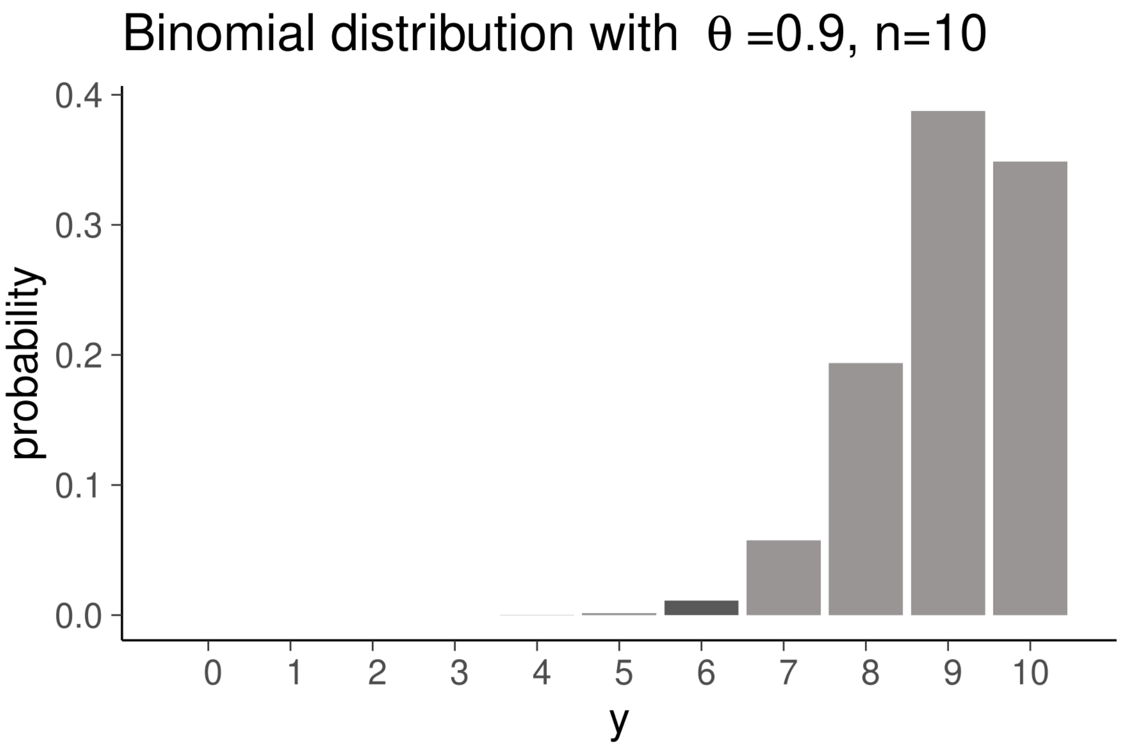
$$p(y|\theta, n) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$



Binomial, what if $y = 9$?

- Observation model (function of y, discrete)

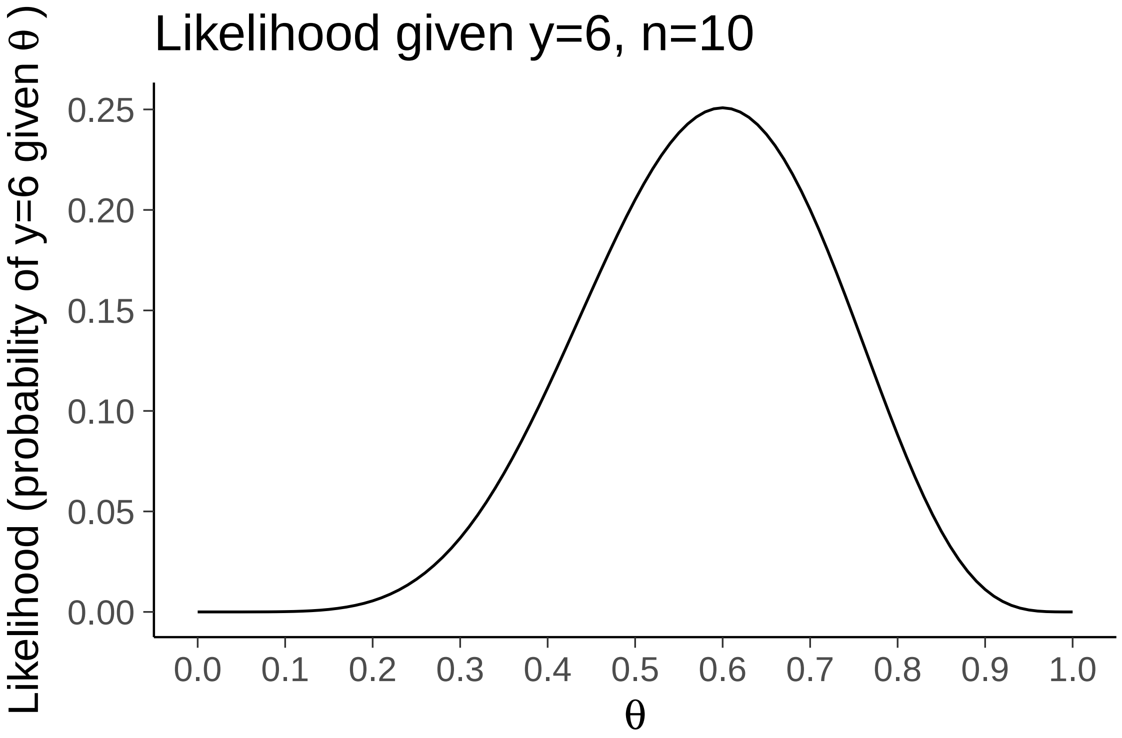
$$p(y|\theta, n) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$



Binomial: unknown θ and $y = 6$

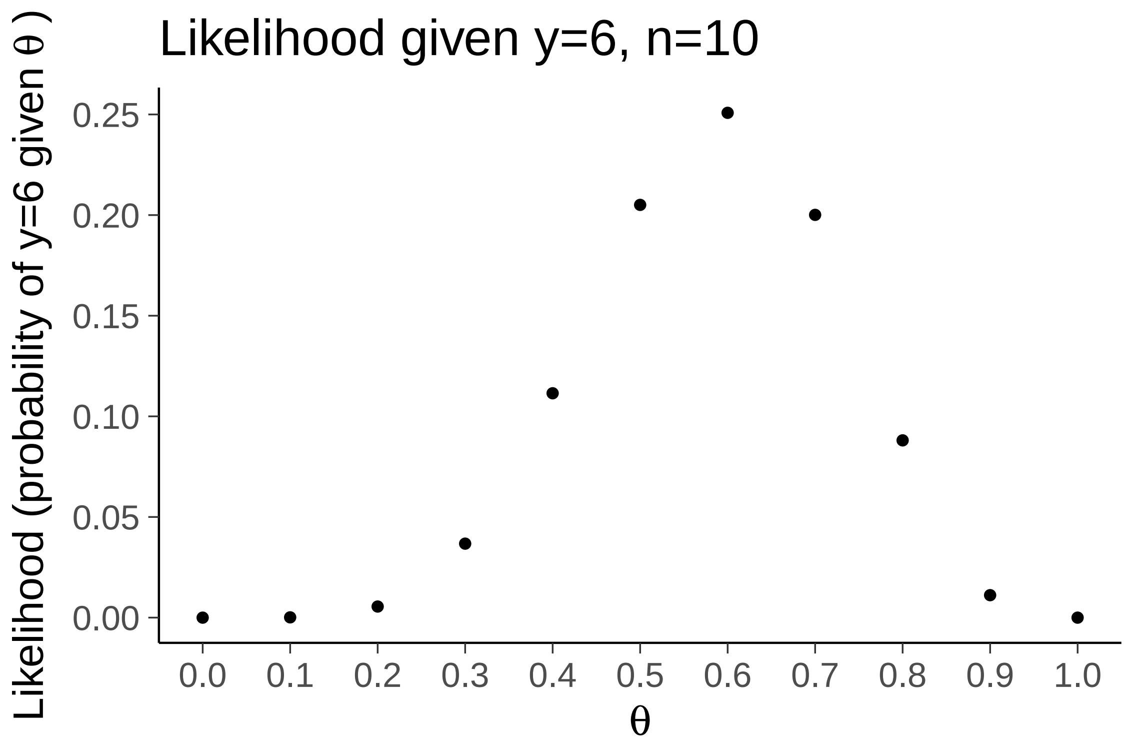
- Likelihood (function of θ, continuous)

$$p(y|\theta, n) = \binom{n}{y} \theta^y (1-\theta)^{n-y}$$



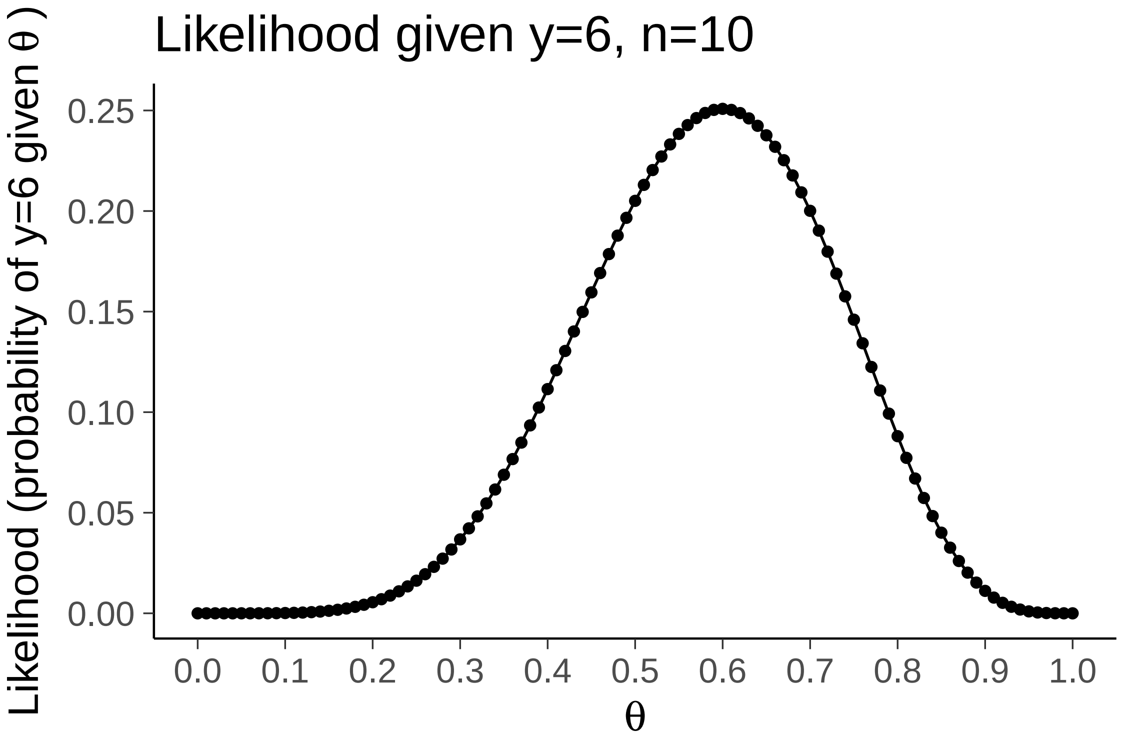
Binomial: unknown θ and $y = 6$

- Theory: We can compute the value for any θ
- Practice: We can only evaluate a finite number of values



Binomial: unknown θ and $y = 6$

- But if we compute enough values and interpolate between them, the plot looks smooth



Binomial: unknown θ and $y = 6$

- Likelihood (function of θ, continuous)

$$p(y|\theta, n) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$$

- likelihood function describes uncertainty, but is not normalized distribution

`integrate(function(theta) dbinom(6, 10, theta), 0, 1)` $\approx 0.09 \neq 1$

Posterior Distribution

- Joint distribution $p(\theta, y \mid M)$
  - M stand for the model, sometimes is omitted
  - factors as $p(\theta, y \mid \mathrm{M}) = p(y|\theta, \mathrm{M})p(\theta|\mathrm{M})$

Posterior Distribution

- Joint distribution $p(\theta, y \mid M)$
  - M stand for the model, sometimes is omitted
  - factors as $p(\theta, y \mid \mathrm{M}) = p(y|\theta, \mathrm{M})p(\theta|\mathrm{M})$

- Posterior with Bayes rule (function of θ, continuous)

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

where $p(y) = \int p(y|\theta)p(\theta)\,\mathrm{d}\theta$

Binomial Posterior

- Start with uniform prior

$$p(\theta) = 1, \quad \text{when} \quad 0 \leq \theta \leq 1$$

Binomial Posterior

- Start with uniform prior

$$p(\theta) = 1, \quad \text{when} \quad 0 \leq \theta \leq 1$$

- Then

$$p(\theta|y) = \frac{p(y|\theta)}{p(y)} = \frac{\binom{n}{y}\theta^y(1-\theta)^{n-y}}{\int_0^1 \binom{n}{y}\theta^y(1-\theta)^{n-y}\,\mathrm{d}\theta}$$

$$= \frac{1}{Z}\theta^y(1-\theta)^{n-y}$$

Binomial Posterior

- Normalization term Z (constant given y)

$$Z = p(y) = \int_0^1 \theta^y (1-\theta)^{n-y}\, \mathrm{d}\theta = \frac{\Gamma(y+1)\Gamma(n-y+1)}{\Gamma(n+2)}$$

Binomial Posterior

- Normalization term Z (constant given y)

$$Z = p(y) = \int_0^1 \theta^y (1-\theta)^{n-y} \,\mathrm{d}\theta = \frac{\Gamma(y+1)\Gamma(n-y+1)}{\Gamma(n+2)}$$

- Evaluate with $y = 6, n = 10$

`y<-6;n<-10;`

`integrate(function(theta) theta^y*(1-theta)^(n-y), 0, 1)` ≈ 0.0004329

`gamma(6+1)*gamma(10-6+1)/gamma(10+2)` ≈ 0.0004329

Binomial Posterior

- Normalization term Z (constant given y)

$$Z = p(y) = \int_0^1 \theta^y (1-\theta)^{n-y} \, \mathrm{d}\theta = \frac{\Gamma(y+1)\Gamma(n-y+1)}{\Gamma(n+2)}$$

- Evaluate with $y = 6, n = 10$

  `y<-6;n<-10;`

  `integrate(function(theta) theta^y*(1-theta)^(n-y), 0, 1)` ≈ 0.0004329

  `gamma(6+1)*gamma(10-6+1)/gamma(10+2)` ≈ 0.0004329

- usually computed via $\log \Gamma(\cdot)$ due to the limitations of floating point representation

Binomial Posterior

- Normalization term in closed form is not in general available
- Later we will learn approximate integration methods that work without knowing the normalization term

Binomial: unknown θ

- Posterior is:

$$p(\theta \mid y) = \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)}\theta^{y}(1-\theta)^{n-y},$$

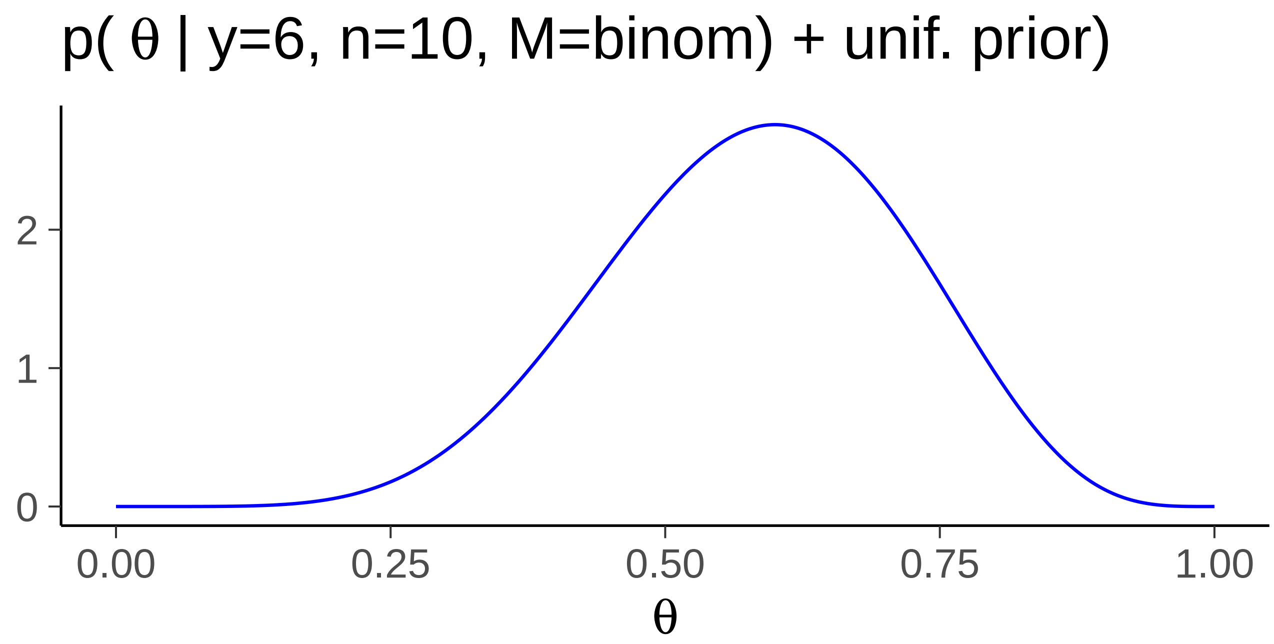
Binomial: unknown θ

- Posterior is:

$$p(\theta \mid y) = \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)}\theta^y(1-\theta)^{n-y},$$

which is called Beta distribution

$$\theta \mid y \sim \mathrm{Beta}(y+1, n-y+1)$$



Binomial: computation (R)

- density `dbeta`
- CDF `pbeta`
- quantile `qbeta`
- random number `rbeta`

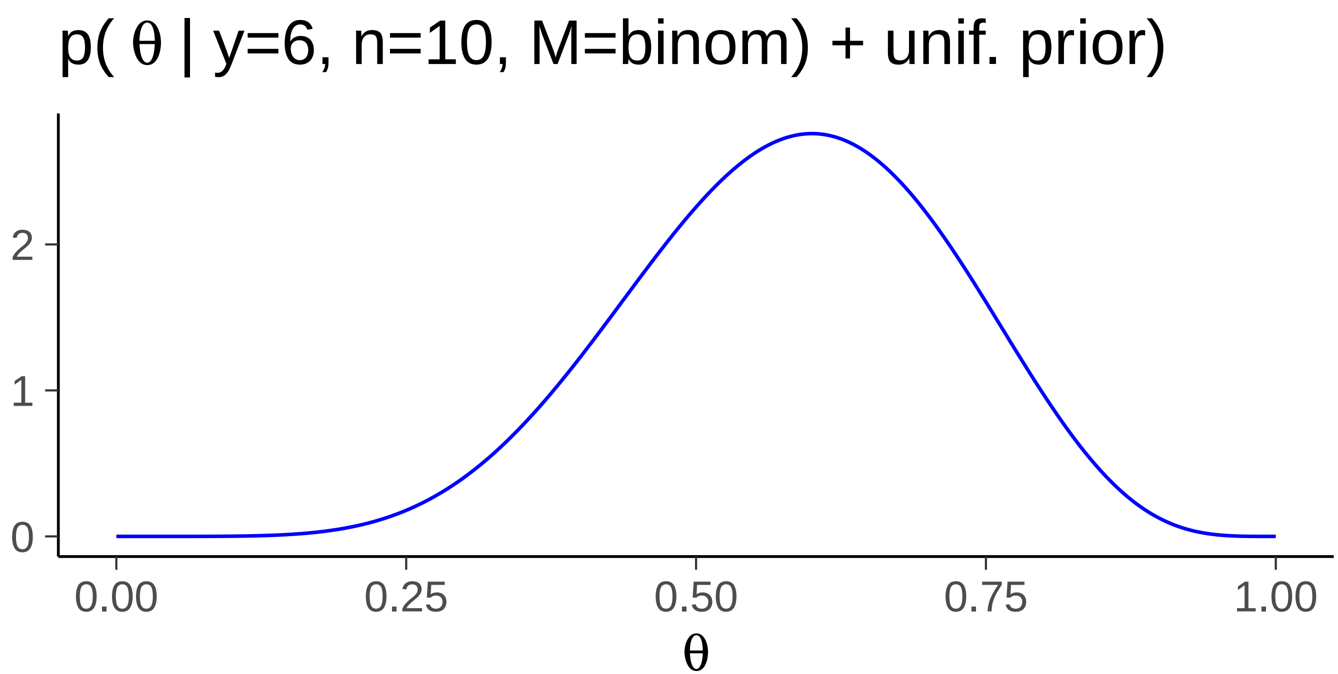
Binomial: computation (Python)

- `from scipy.stats import beta`
- density `beta.pdf`
- CDF `beta.cdf`
- quantile `beta.ppf`
- random number `beta.rvs`

- `from preliz import Beta`
- Similar to scipy.stats
- Nicer features to explore distributions
- https://preliz.readthedocs.io

Binomial: computation

- No simple formula exists for the Beta CDF
- Modern software use sophisticated numerical methods
- Over 200 years ago, Laplace hit the same wall, so he invented the normal approximation (Laplace approximation, BDA3 Ch. 4)

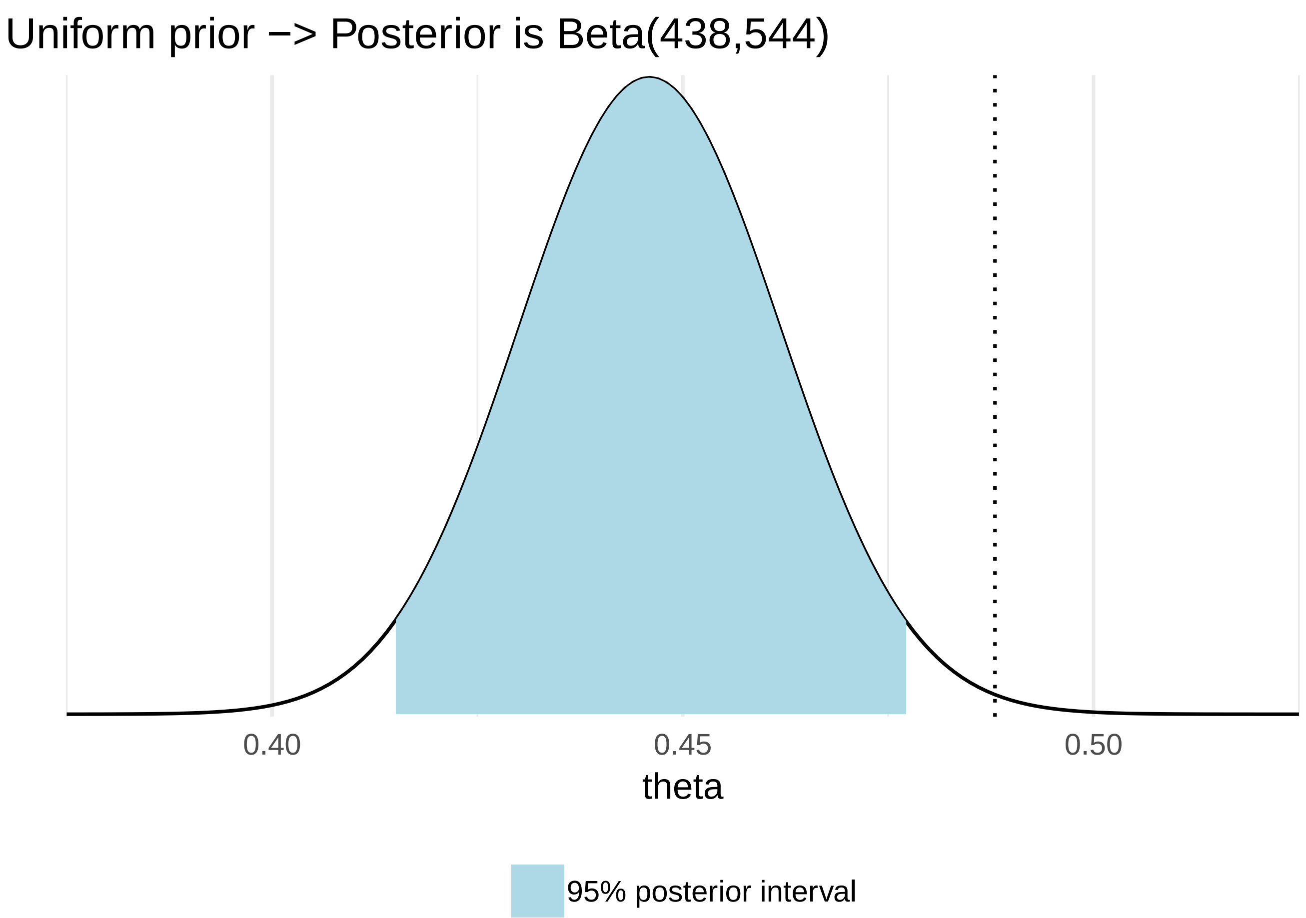


Placenta previa

- Probability of a girl birth given placenta previa (BDA3 p. 37)
  - 437 girls and 543 boys have been observed
  - is the proportion of girls 0.445 different from the population average 0.485?

Placenta previa

- Probability of a girl birth given placenta previa (BDA3 p. 37)
  - 437 girls and 543 boys have been observed
  - is the proportion of girls 0.445 different from the population average 0.485?



Predictive distribution – Effect of integration

- Predictive distribution for new $\tilde{y}$ (discrete)

$$
\begin{aligned}
p(\tilde{y}{=}1|y,M) &= \int_0^1 \underbrace{p(\tilde{y}{=}1|\theta,y,M)}_{\theta}\, p(\theta|y,M)\,\mathrm{d}\theta \\
&= \int_0^1 \theta p(\theta|y,M)\,\mathrm{d}\theta \\
&= \mathbb{E}[\theta|y]
\end{aligned}
$$

Predictive distribution – Effect of integration

- With uniform prior

$$\mathbb{E}[\theta|y] = \frac{y+1}{n+2}$$

- Extreme cases

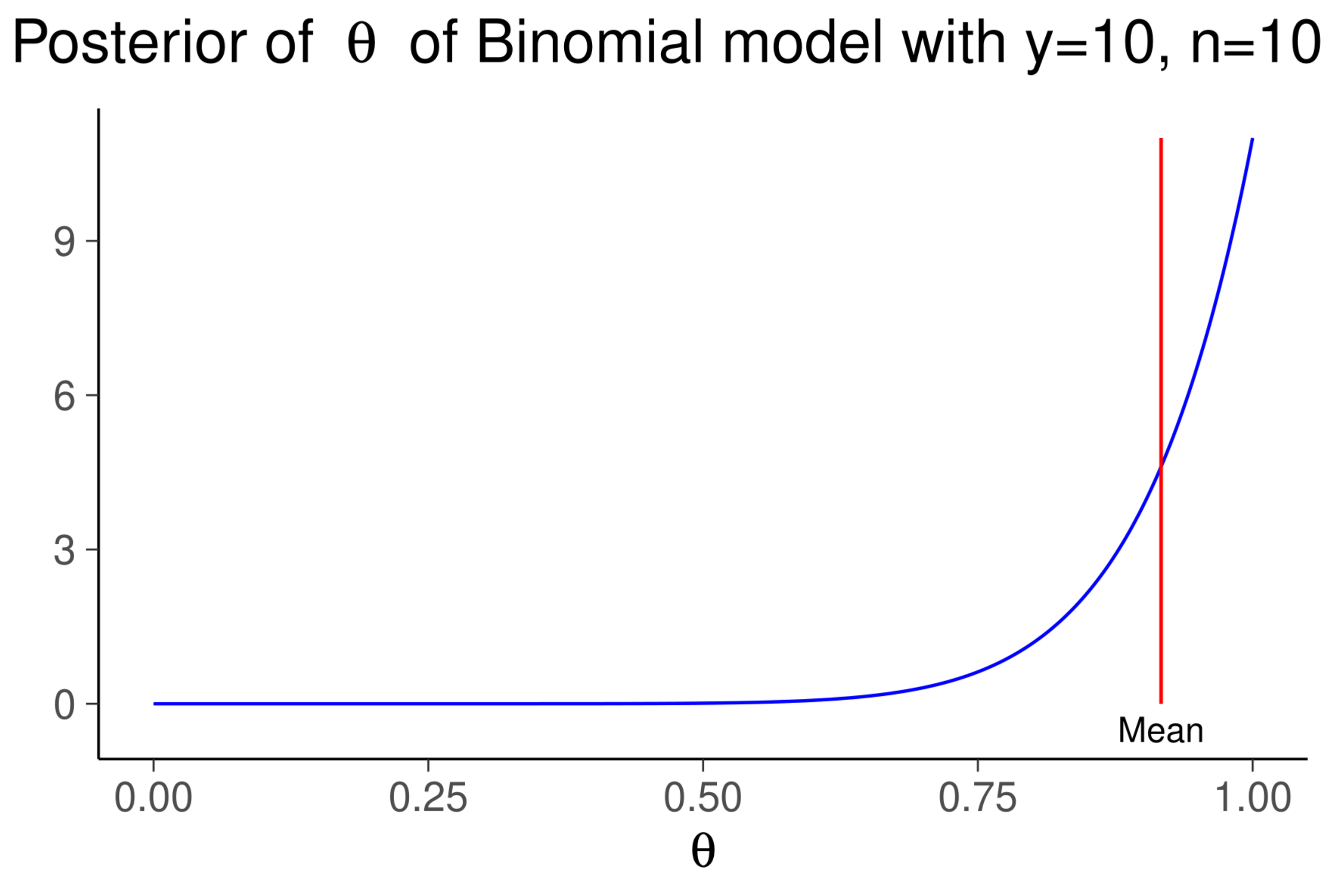
$$p(\tilde{y} = 1|y = 0) = \frac{1}{n+2}$$
$$p(\tilde{y} = 1|y = n) = \frac{n+1}{n+2}$$

- cf. maximum likelihood

$$p(\tilde{y} = 1|y = 0) = 0$$
$$p(\tilde{y} = 1|y = n) = 1$$

Benefits of integration

Example: $n = 10, y = 10$



Predictive distribution

- Prior predictive distribution for new $\tilde{y}$ (discrete)

$$p(\tilde{y} = 1|\ M) = \int_0^1 p(\tilde{y} = 1|\ \theta, M)\ \mathrm{p(theta \mid M)}\ d\theta$$

- Posterior predictive distribution for new $\tilde{y}$ (discrete)

$$p(\tilde{y} = 1|\ y, M) = \int_0^1 p(\tilde{y} = 1|\ \theta, y, M)\ \mathrm{p(theta \mid y,\ M)}\ d\theta$$

Left handedness

- If we would like to provide scissors for all students, how many left handed scissors we would need?

Left handedness

- If we would like to provide scissors for all students, how many left handed scissors we would need?
- Posterior distribution for θ is

$$\mathrm{Beta}(\alpha + l, \beta + r)$$

Left handedness

- If we would like to provide scissors for all students, how many left handed scissors we would need?
- Posterior distribution for θ is

$$\mathrm{Beta}(\alpha + l, \beta + r)$$

- Posterior predictive distribution for $\tilde{l}$ is:

$$\text{Beta-Binomial}\big(\tilde{l} \mid N - n, \alpha + l, \beta + r\big)$$

Left handedness

- If we would like to provide scissors for all students, how many left handed scissors we would need?
- Posterior distribution for θ is

$$\mathrm{Beta}(\alpha + l, \beta + r)$$

- Posterior predictive distribution for $\tilde{l}$ is:

$$\text{Beta-Binomial}(\tilde{l} \mid N - n, \alpha + l, \beta + r)$$

- Demo: https://handedness.streamlit.app

Priors

- Conjugate prior (BDA3 p. 35)
- Noninformative prior (BDA3 p. 51)
- Proper and improper prior (BDA3 p. 52)
- Weakly informative prior (BDA3 p. 55)
- Informative prior (BDA3 p. 55)
- Prior sensitivity (BDA3 p. 38)

Conjugate prior

- Prior and posterior have the same form
  - only for a few cases like exponential family distributions
- Used to be important for computational reasons, and still sometimes used for special models to allow partial analytic marginalization (Ch 3)
  - with modern sampling methods like Hamiltonian Monte Carlo / NUTS no computational benefit

Beta prior for Binomial model

- Prior

$$\mathrm{Beta}(\theta \mid \alpha, \beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

Beta prior for Binomial model

- Prior

$$\mathrm{Beta}(\theta \mid \alpha, \beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

- Posterior

$$\begin{aligned} p(\theta \mid y, M) &\propto \theta^{y}(1-\theta)^{n-y}\, \theta^{\alpha-1}(1-\theta)^{\beta-1} \\ &\propto \theta^{y+\alpha-1}(1-\theta)^{n-y+\beta-1} \end{aligned}$$

after normalization

$$p(\theta \mid y, M) = \mathrm{Beta}(\theta \mid \alpha+y, \beta+n-y)$$

Beta prior for Binomial model

- Prior

$$\mathrm{Beta}(\theta \mid \alpha, \beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

- Posterior

$$\begin{aligned} p(\theta \mid y, M) &\propto \theta^{y}(1-\theta)^{n-y}\, \theta^{\alpha-1}(1-\theta)^{\beta-1} \\ &\propto \theta^{y+\alpha-1}(1-\theta)^{n-y+\beta-1} \end{aligned}$$

  after normalization

$$p(\theta \mid y, M) = \mathrm{Beta}(\theta \mid \alpha+y, \beta+n-y)$$

- $(\alpha-1)$ and $(\beta-1)$ can be considered to be the number of prior observations

Beta prior for Binomial model

- Prior

$$\mathrm{Beta}(\theta \mid \alpha, \beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

- Posterior

$$\begin{aligned} p(\theta \mid y, M) &\propto \theta^{y}(1-\theta)^{n-y}\, \theta^{\alpha-1}(1-\theta)^{\beta-1} \\ &\propto \theta^{y+\alpha-1}(1-\theta)^{n-y+\beta-1} \end{aligned}$$

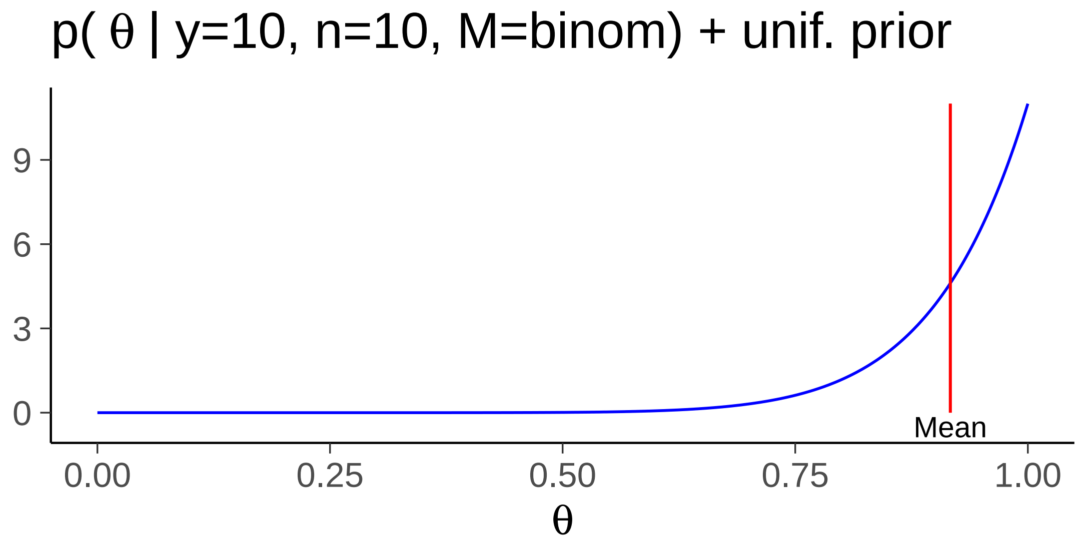
  after normalization

$$p(\theta \mid y, M) = \mathrm{Beta}(\theta \mid \alpha + y, \beta + n - y)$$

- $(\alpha - 1)$ and $(\beta - 1)$ can be considered to be the number of prior observations
- Uniform prior when $\alpha = 1$ and $\beta = 1$

Benefits of integration and prior

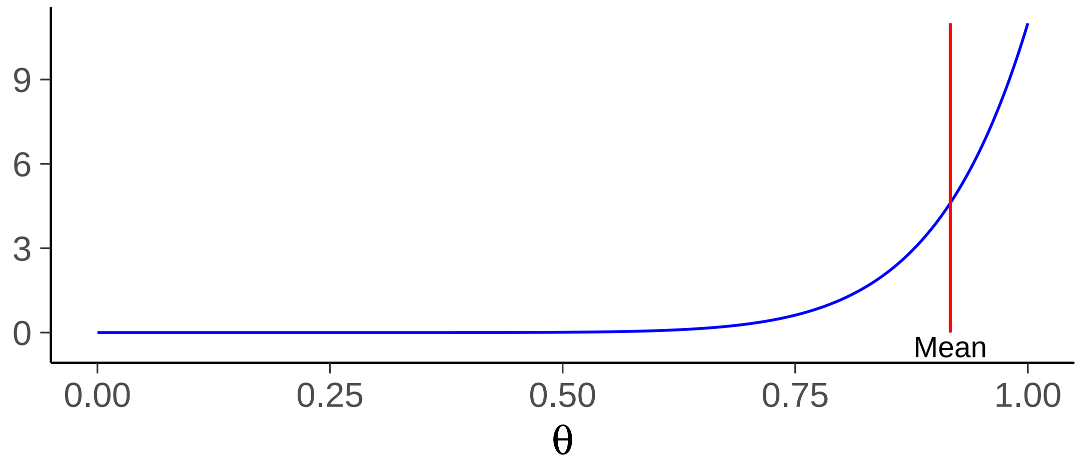
Example: $n = 10, y = 10$ – uniform vs Beta(2,2) prior



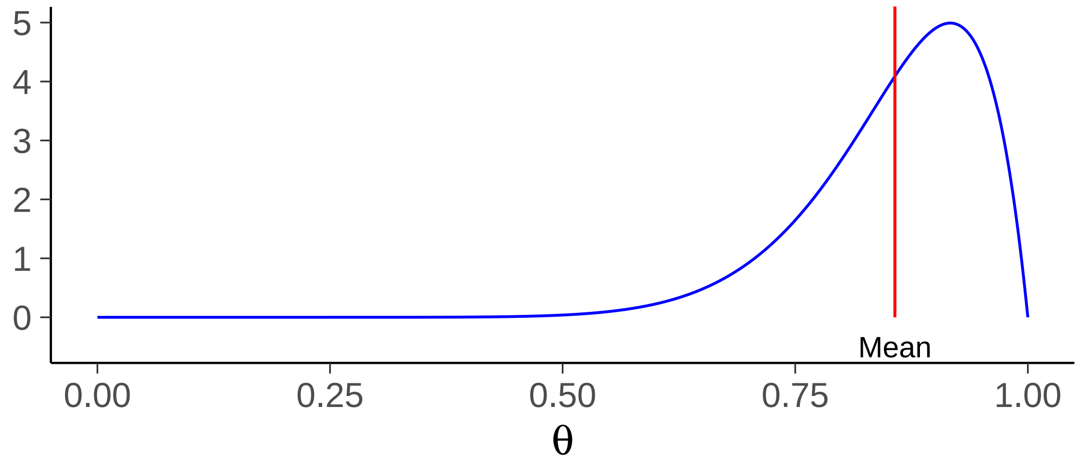
Benefits of integration and prior

Example: $n = 10, y = 10$ – uniform vs Beta(2,2) prior

p(θ | y=10, n=10, M=binom) + unif. prior



p(θ | y=10, n=10, M=binom) + Beta(2,2) prior



Beta prior for Binomial model

- Posterior

$$p(\theta \mid y, M) = \mathrm{Beta}(\theta \mid \alpha + y, \beta + n - y)$$

Beta prior for Binomial model

- Posterior

$$p(\theta \mid y, M) = \mathrm{Beta}(\theta \mid \alpha + y, \beta + n - y)$$

- Posterior mean

$$E[\theta \mid y] = \frac{\alpha + y}{\alpha + \beta + n}$$

  - combination of prior and likelihood information
  - when $n \to \infty$, $E[\theta \mid y] \to \frac{y}{n}$

Beta prior for Binomial model

- Posterior

$$p(\theta \mid y, M) = \mathrm{Beta}(\theta \mid \alpha + y, \beta + n - y)$$

- Posterior mean

$$E[\theta \mid y] = \frac{\alpha + y}{\alpha + \beta + n}$$

  - combination of prior and likelihood information
  - when $n \to \infty$, $E[\theta \mid y] \to \frac{y}{n}$
- Posterior variance

$$\mathrm{Var}[\theta \mid y] = \frac{E[\theta \mid y](1 - E[\theta \mid y])}{\alpha + \beta + n + 1}$$

  - decreases when n increases
  - when $n \to \infty$, $\mathrm{Var}[\theta \mid y] \to 0$

Noninformative prior, proper and improper prior

- Vague, flat, diffuse, or noninformative
  - try to “let the data speak for themselves”
  - flat is not non-informative
  - flat can be stupid
  - making prior flat somewhere can make it non-flat somewhere else

- Proper prior has $\int p(\theta) = 1$
- Improper prior density doesn’t have a finite integral
  - the posterior can still sometimes be proper

Weakly informative priors

- Weakly informative priors produce computationally better behaving posteriors
  - quite often there's at least some knowledge about the scale
  - useful also if there's more information from previous observations, but not certain how well that information is applicable in a new case

Weakly informative priors

- Weakly informative priors produce computationally better behaving posteriors
  - quite often there's at least some knowledge about the scale
  - useful also if there's more information from previous observations, but not certain how well that information is applicable in a new case
- Construction
  - Start with some version of a noninformative prior distribution and then add enough information so that inferences are constrained to be reasonable.
  - Start with a strong, highly informative prior and broaden it to account for uncertainty in one's prior beliefs and in the applicability of any historically based prior distribution to new data.

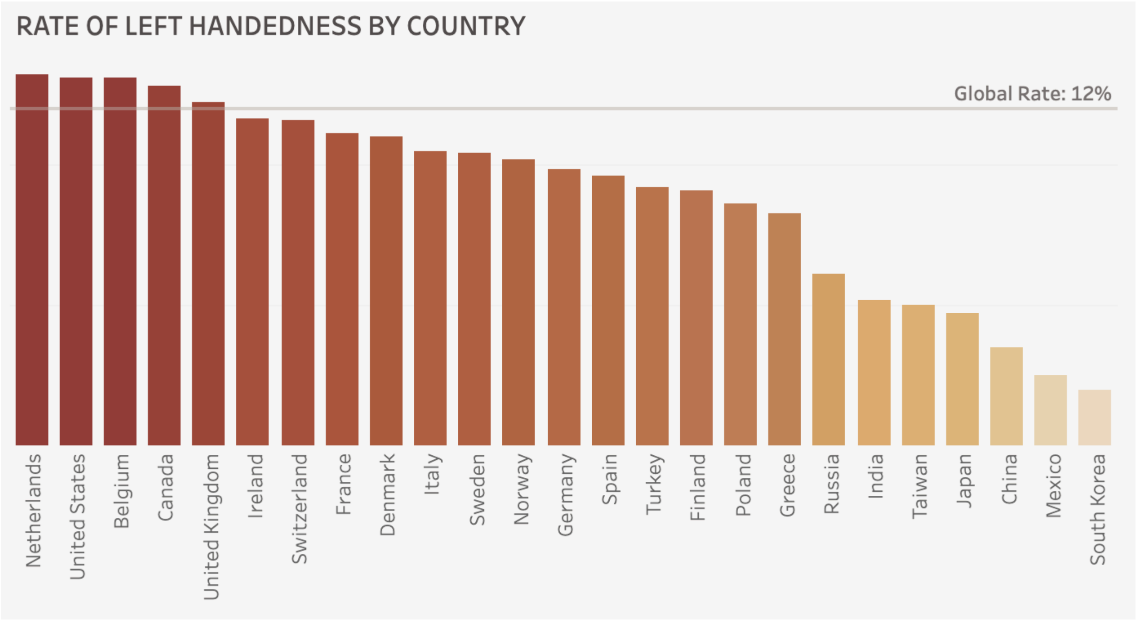
Weakly informative priors

- Weakly informative priors produce computationally better behaving posteriors
  - quite often there's at least some knowledge about the scale
  - useful also if there's more information from previous observations, but not certain how well that information is applicable in a new case
- Construction
  - Start with some version of a noninformative prior distribution and then add enough information so that inferences are constrained to be reasonable.
  - Start with a strong, highly informative prior and broaden it to account for uncertainty in one's prior beliefs and in the applicability of any historically based prior distribution to new data.
- Stan team prior choice recommendations https://github.com/stan-dev/stan/wiki/Prior-Choice-Recommendations

Informative prior for left handedness

- Papadatou-Pastou et al. (2020). Human handedness: A meta-analysis. Psychological Bulletin, 146(6), 481–524. https://doi.org/10.1037/bul0000229
  - totaling 2,396,170 individuals
  - varies between 9.3% and 18.1%, depending on how handedness is measured
  - varies between countries and in time

Informative prior for left handedness



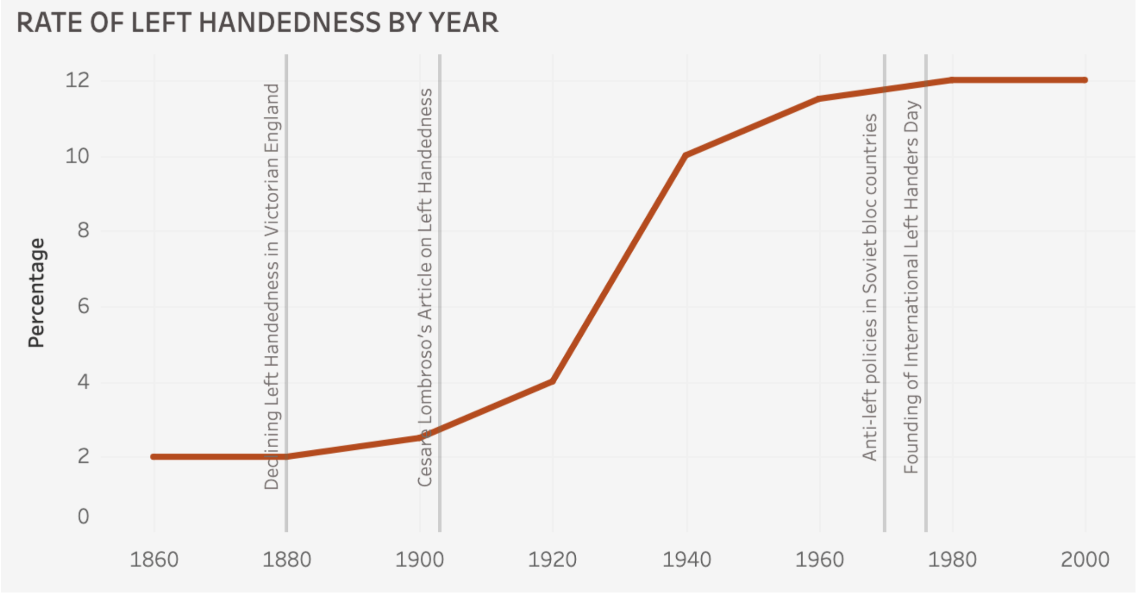
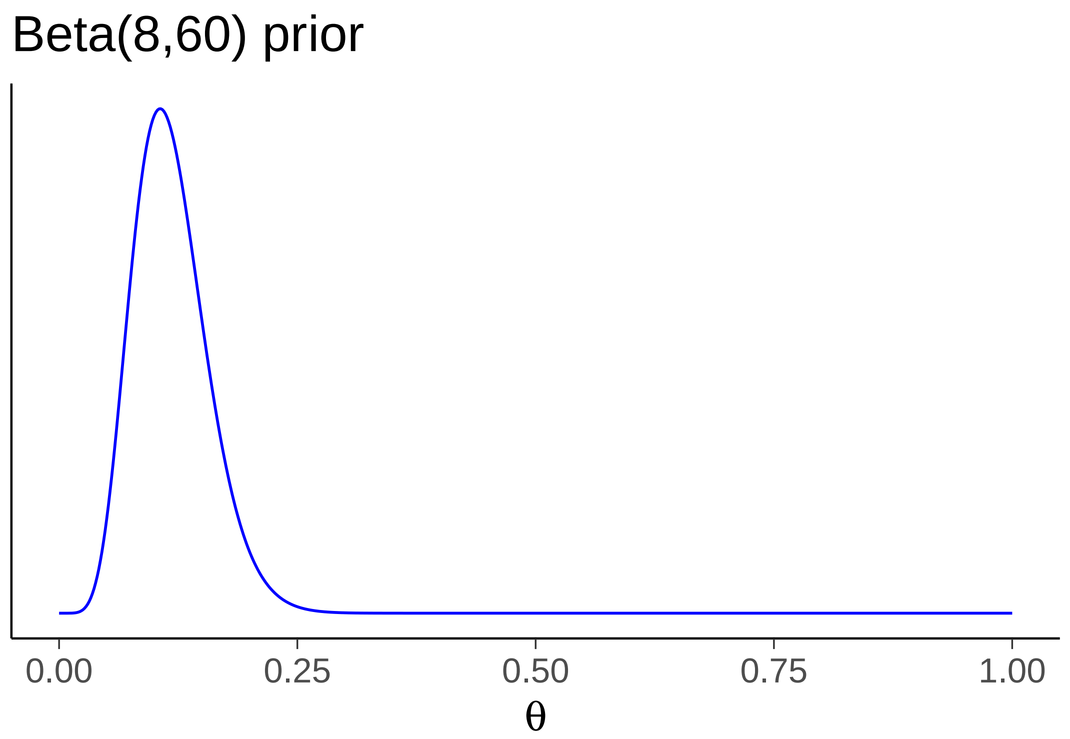


Fig from https://www.reddit.com/r/dataisbeautiful/comments/s9×1ya/history_of_lefthandedness_oc/

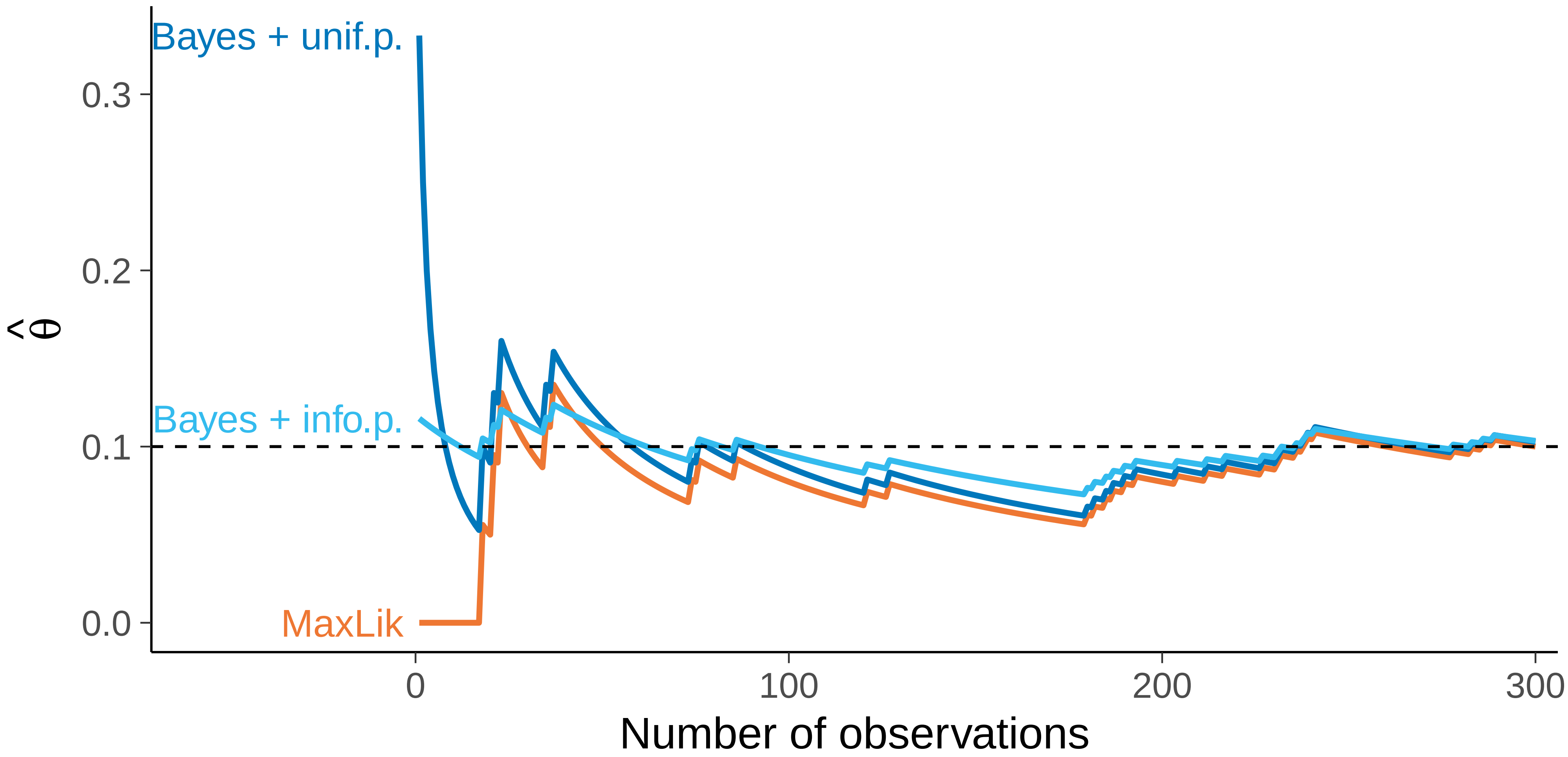
Informative prior for left handedness



Demo: https://handedness.streamlit.app

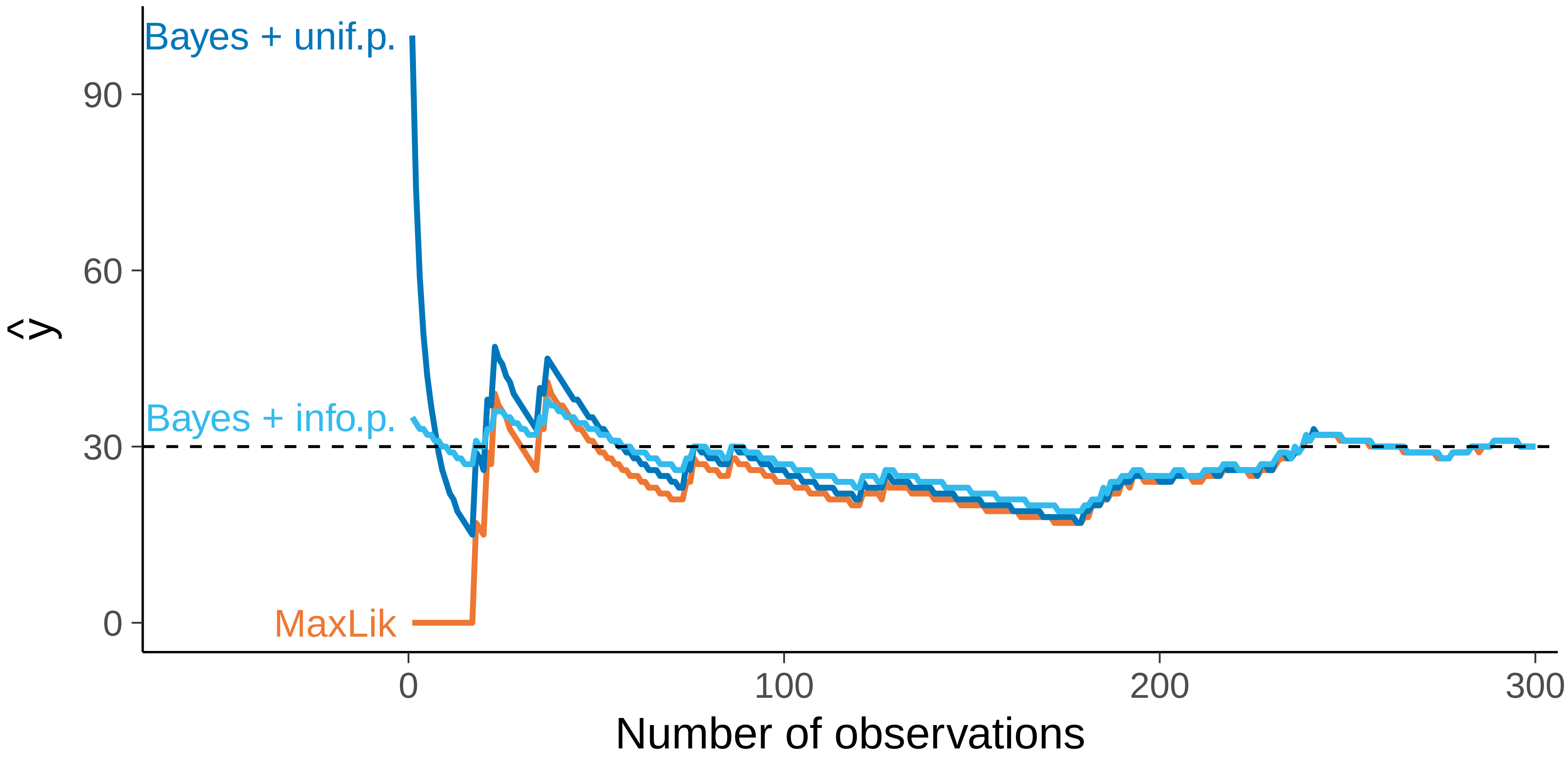
Benefits of integration and prior

- Left handed simulation with $L = 30$ left handed and $N = 300$ total



Benefits of integration and prior

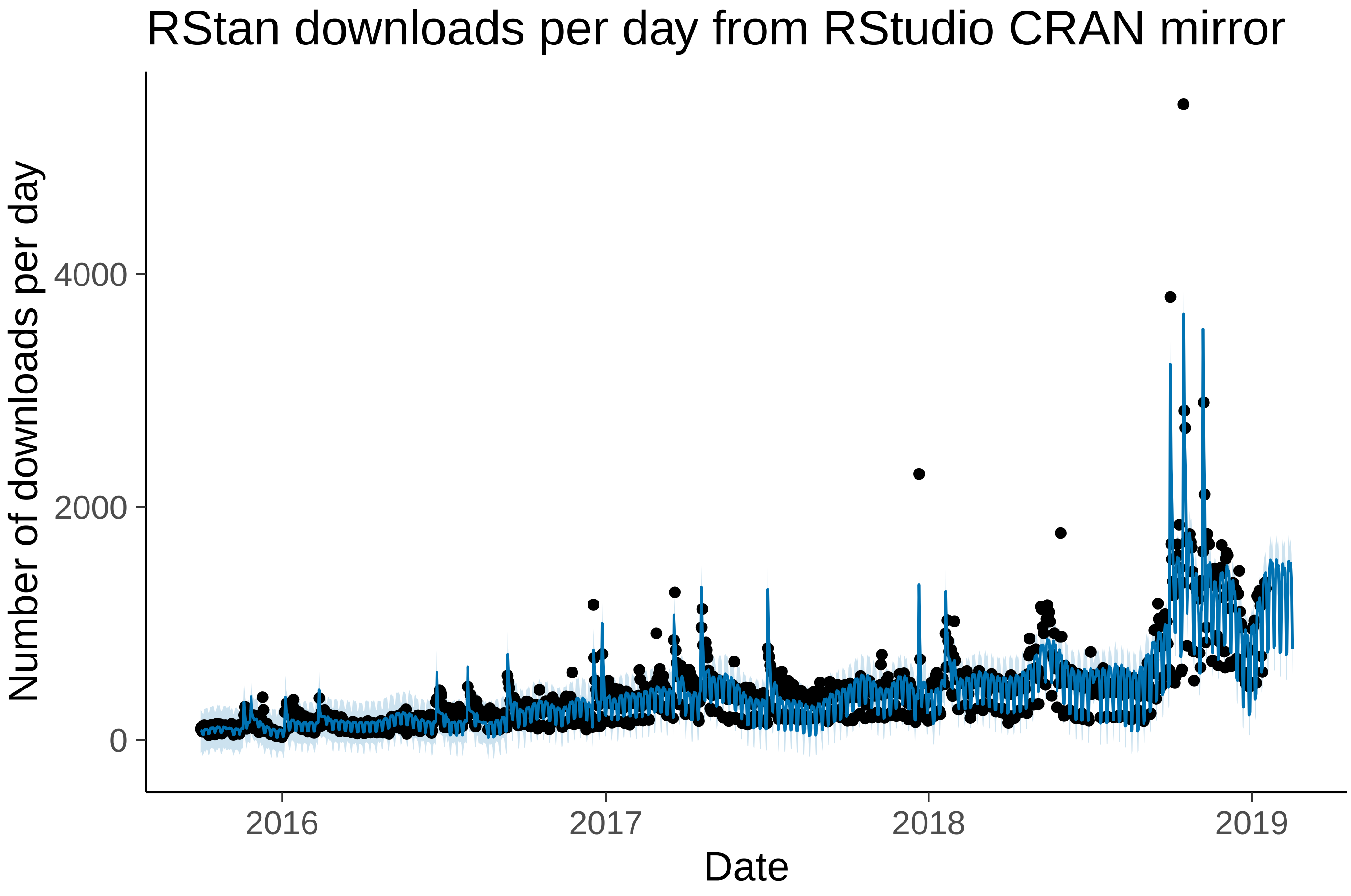
- Left handed simulation with $L = 30$ left handed and $N = 300$ total



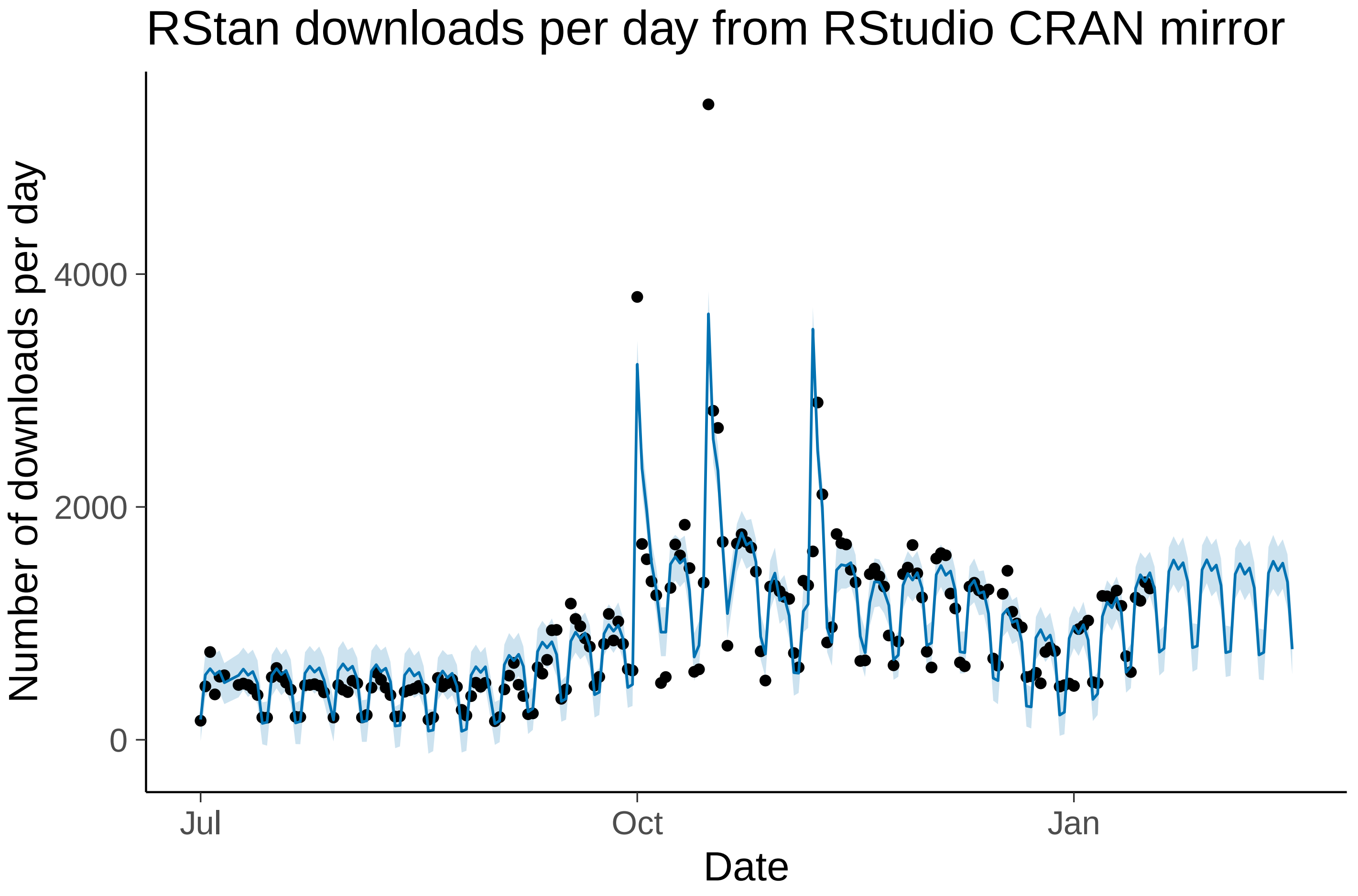
But what if my prior is wrong?

- Introduce bias, but often still produce smaller estimation error because the variance is reduced
  - bias-variance tradeoff

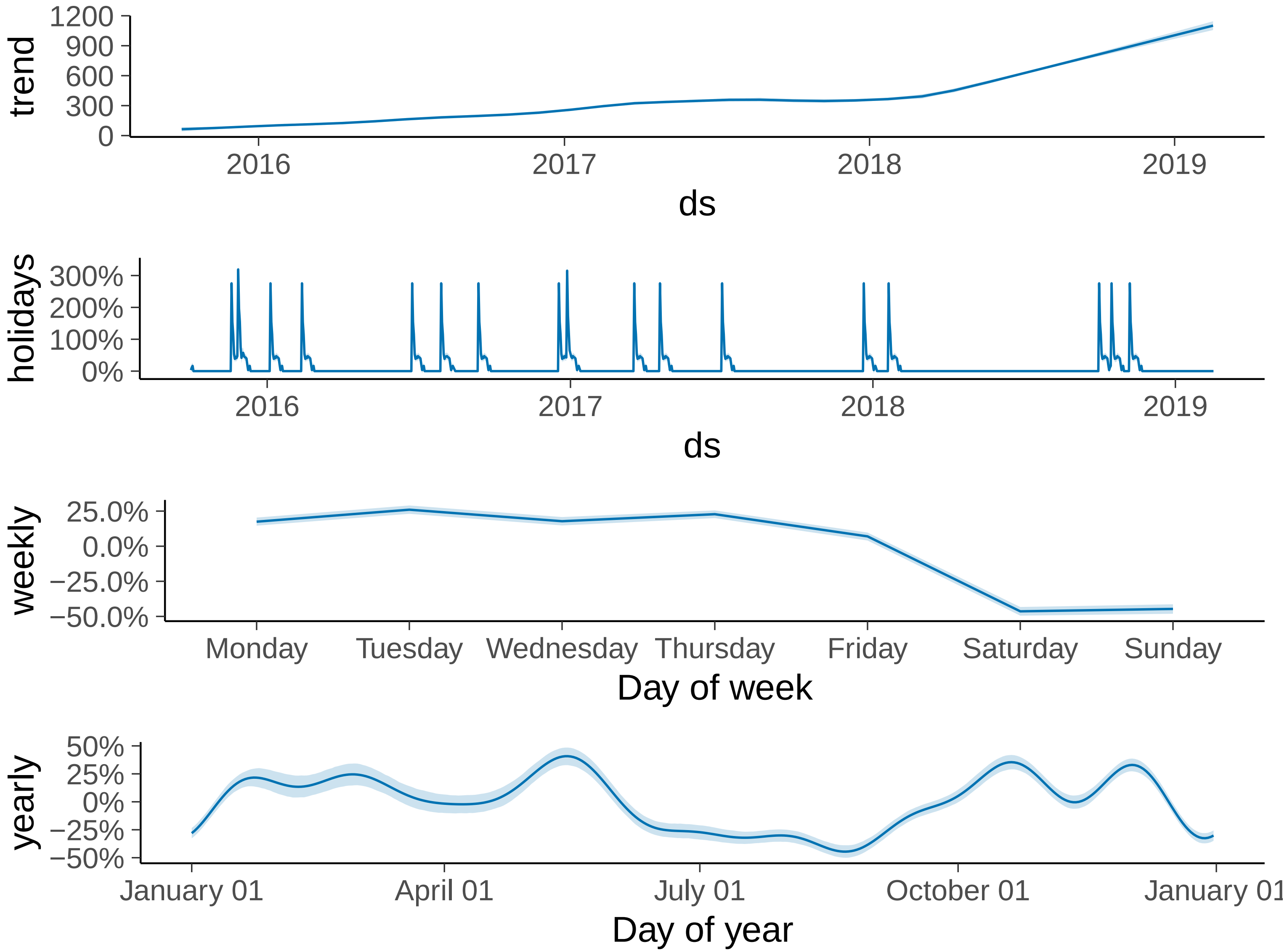
Structural information in predicting future



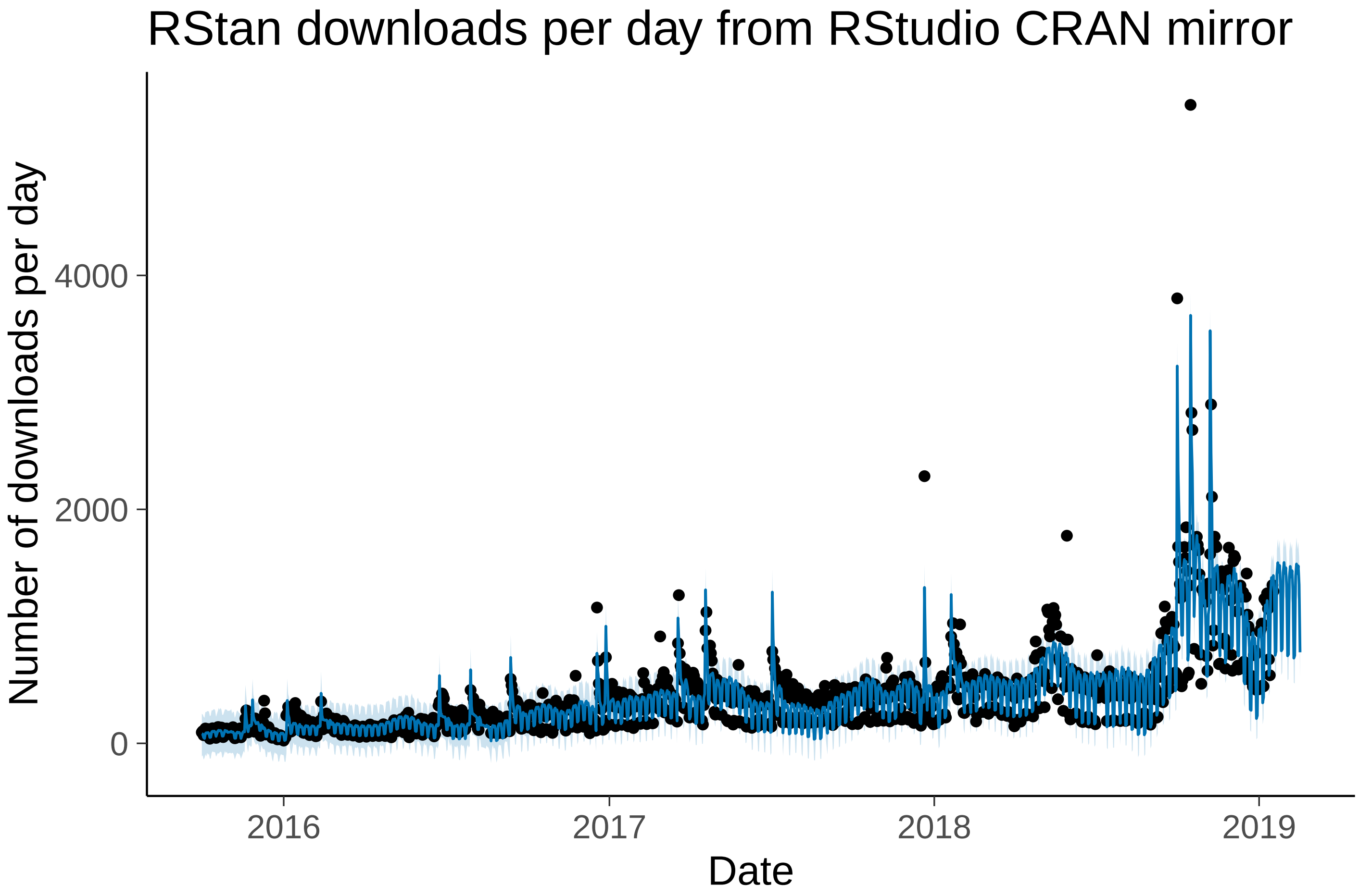
Structural information in predicting future



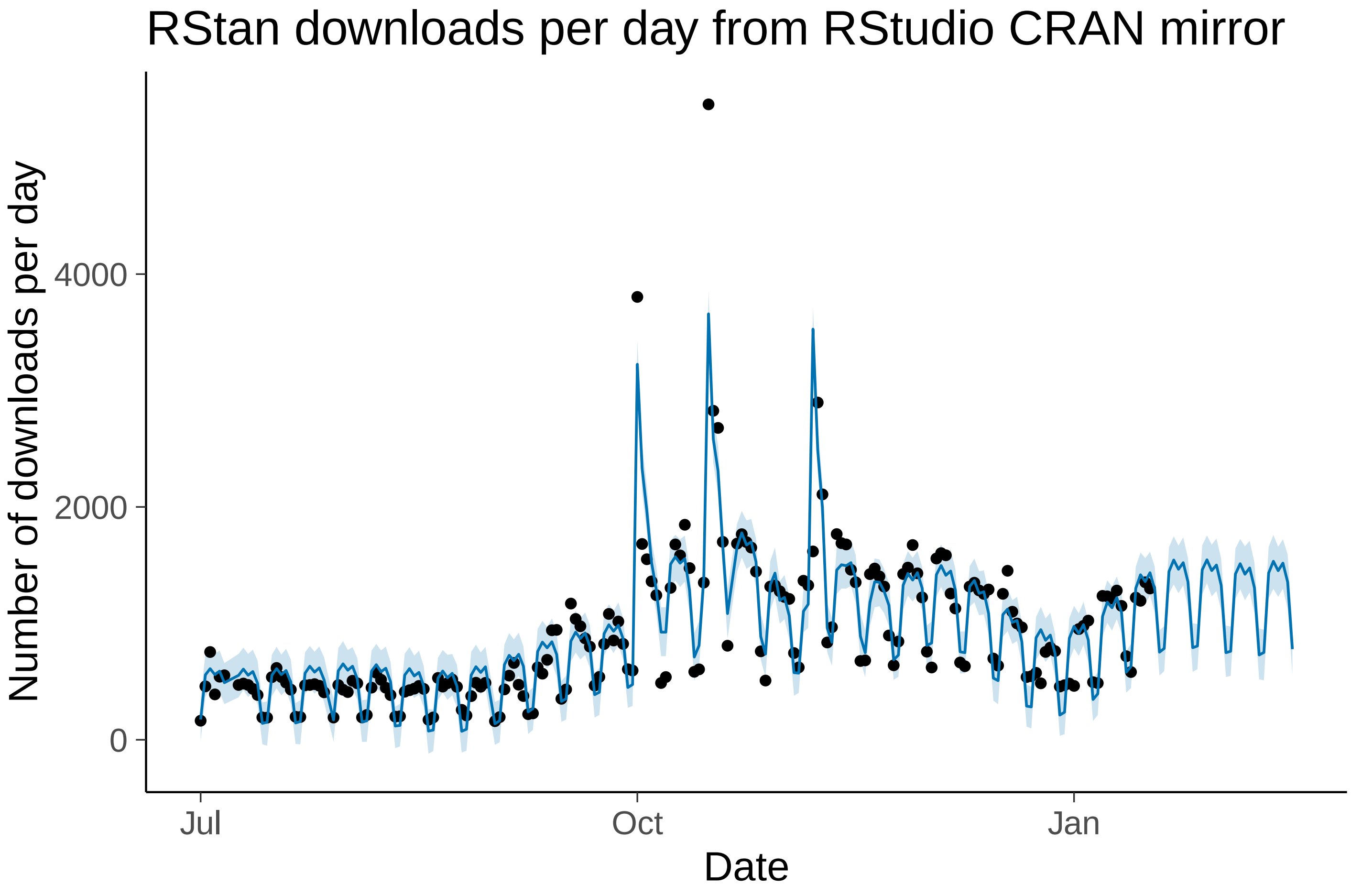
Structural information in predicting future



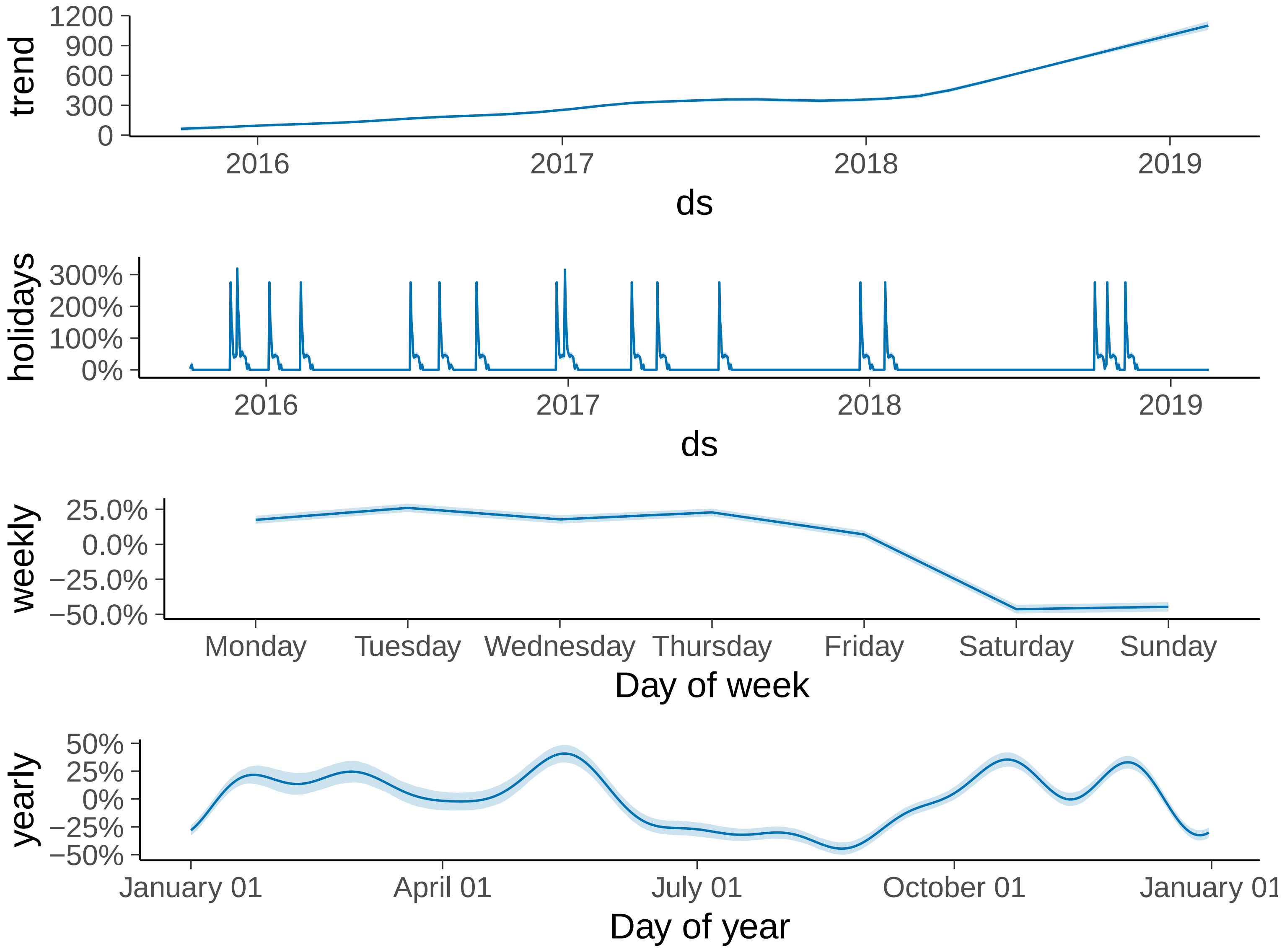
Structural information Prophet by Facebook



Structural information Prophet by Facebook



Structural information Prophet by Facebook



Sufficient statistics

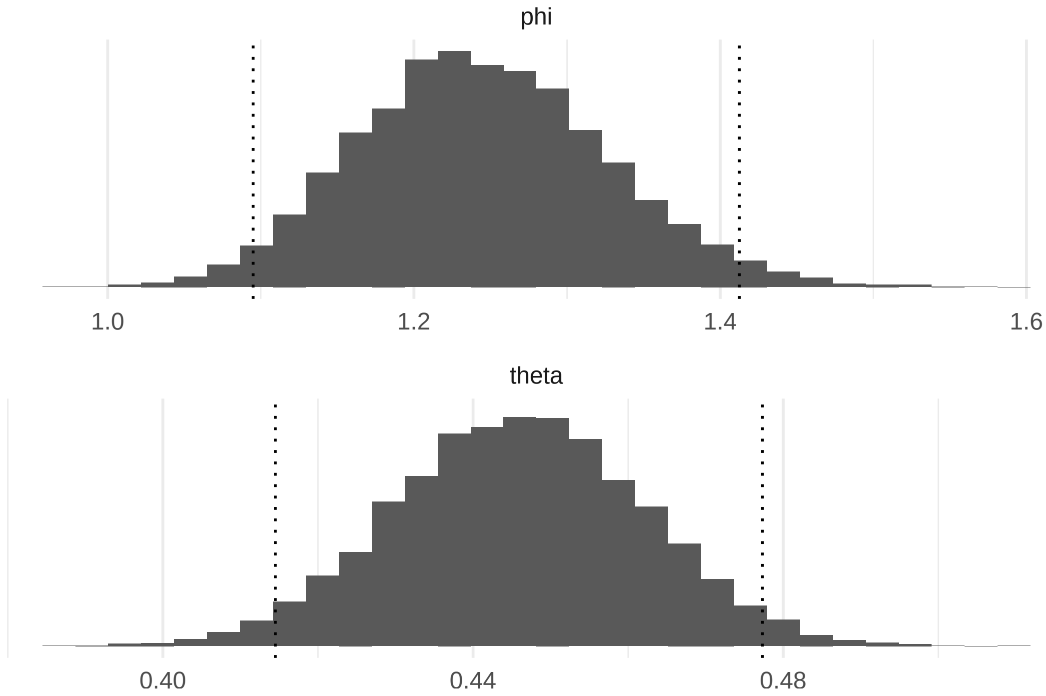
- The quantity $t(y)$ is said to be a *sufficient statistic* for θ, because the likelihood for θ depends on the data y only through the value of $t(y)$.

Sufficient statistics

- The quantity $t(y)$ is said to be a *sufficient statistic* for θ, because the likelihood for θ depends on the data y only through the value of $t(y)$.
- For binomial model the sufficient statistics are y and n (the order doesn't matter)

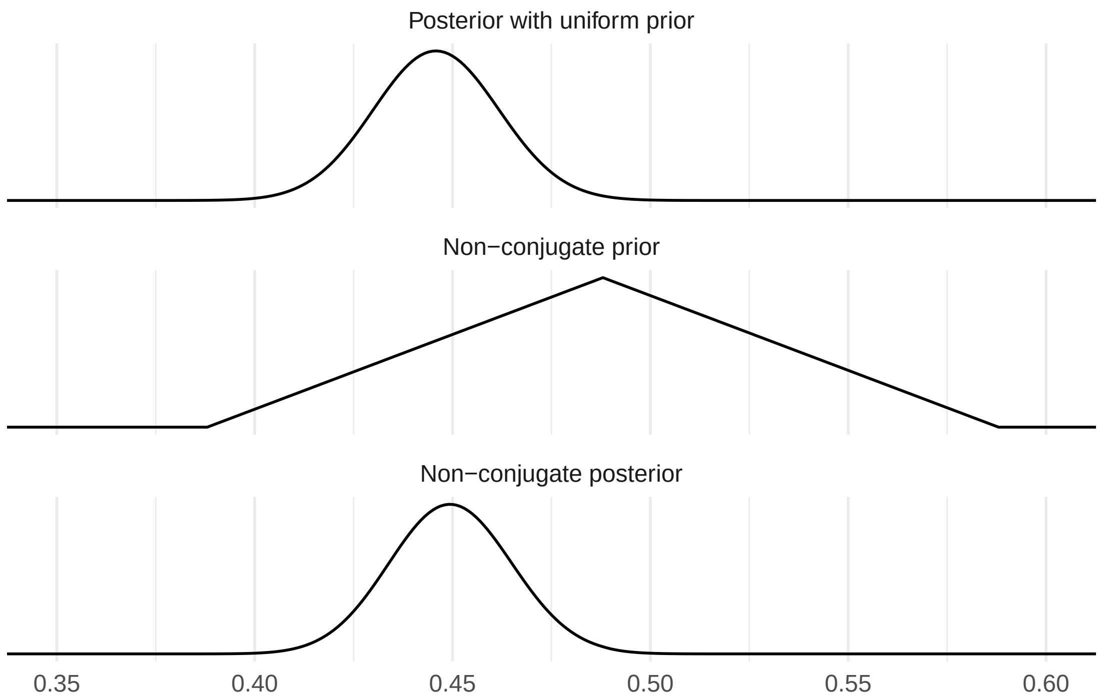
Posterior visualization and inference demos

- demo2_3: Simulate samples from Beta(438, 544), and draw a histogram of θ with quantiles.



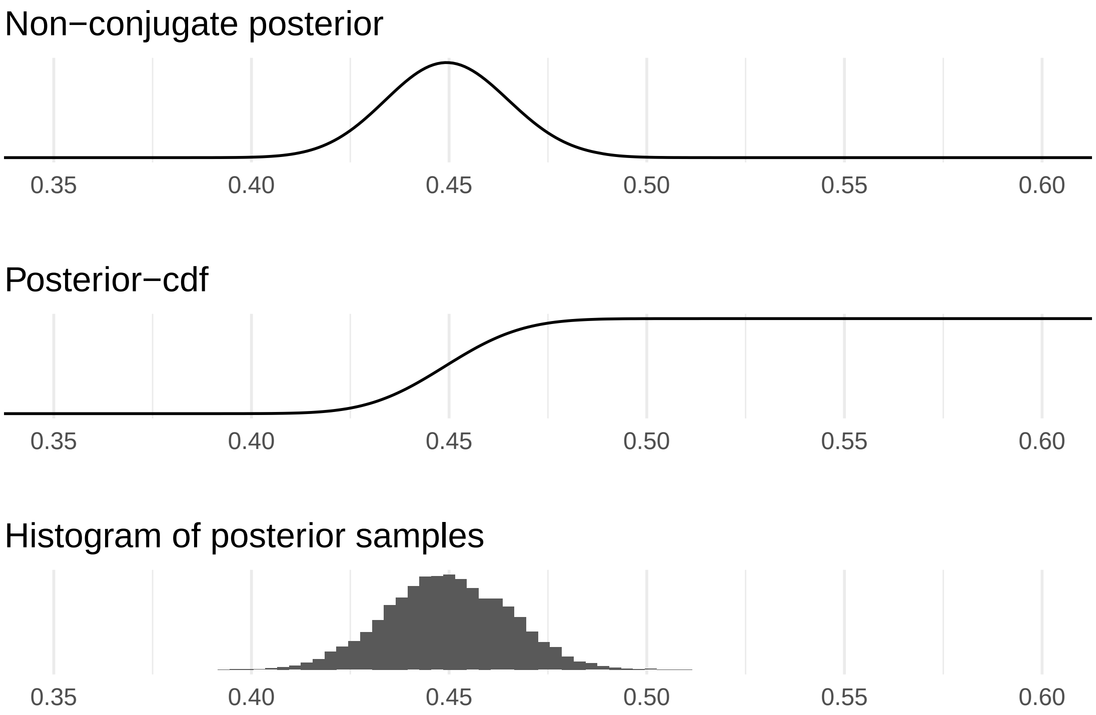
Posterior visualization and inference demos

- demo2_4: Compute posterior distribution in a grid.



Posterior visualization and inference demos

- demo2_4: Sample using the inverse-cdf method.



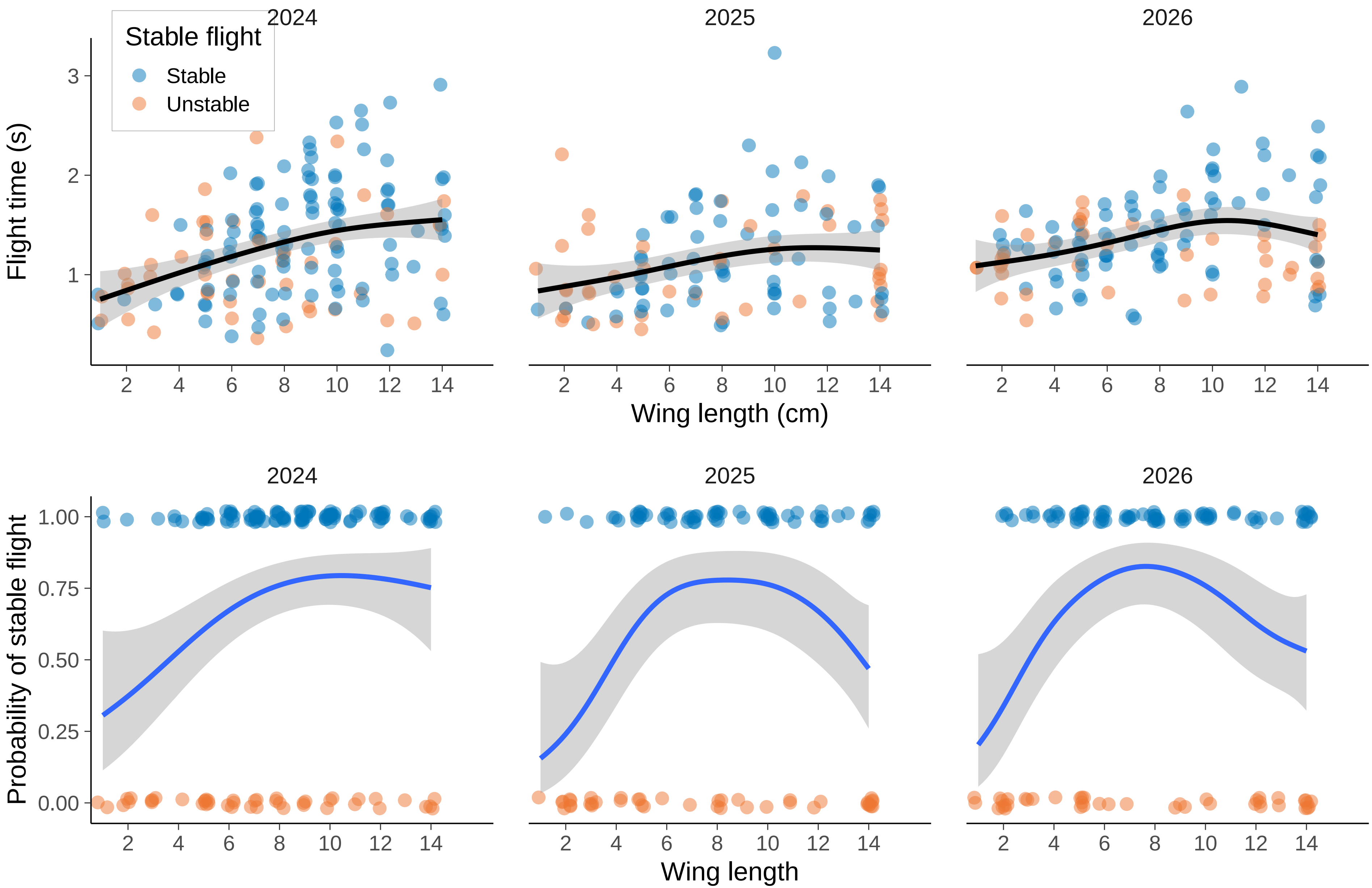
Algae

Algae status is monitored in 274 sites at Finnish lakes and rivers. The observations for the 2008 algae status at each site are presented in file algae.(rda|txt) *('0': no algae, '1': algae present).* Let θ be the probability of a monitoring site having detectable blue-green algae levels.

- Use a binomial model for observations and a $\mathrm{Beta}(2, 10)$ prior.
- What can you say about the value of the unknown θ?
- Experiment how the result changes if you change the prior.

Binomial model with $\theta = f(x)$

- Next week you learn how the binomial model parameter θ can depend on some other measurement x

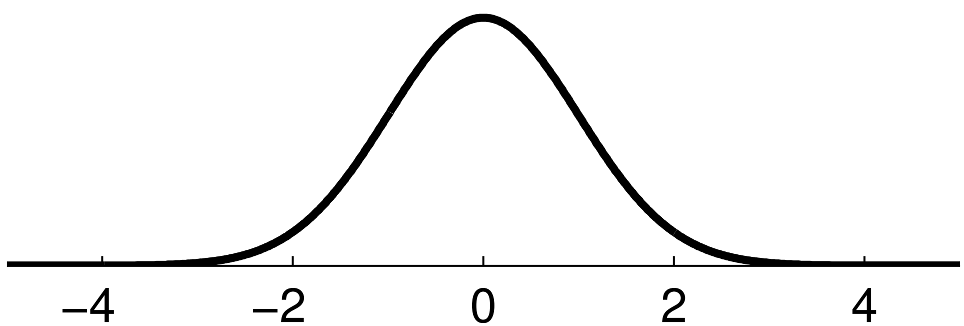


Normal / Gaussian

- Observations y real valued
- Mean θ and variance σ^2 (or standard deviation σ)

$$p(y \mid \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\left(\frac{1}{2\sigma^2}\right)(y-\theta)^2\right)$$

$$y \sim N(\theta, \sigma^2)$$



Reasons to use Normal distribution

- Computational convenience
- Tradition
- Often good approximation (justification based on central limit theorem)

Central limit theorem*

- Given certain conditions, distribution of sum (and mean) of random variables approach Gaussian distribution as $n \to \infty$

Central limit theorem*

- Given certain conditions, distribution of sum (and mean) of random variables approach Gaussian distribution as $n \to \infty$
- Problems
  - does not hold for distributions with infinite variance, e.g., Cauchy
  - may require large n, e.g. Binomial, when θ close to 0 or 1
  - does not hold if one of the variables has much larger scale

Normal distribution – conjugate prior for θ

- Assume σ^2 known

Likelihood $$p(y \mid \theta) \propto \exp\left(-\left(\frac{1}{2\sigma^2}\right)(y-\theta)^2\right)$$

Prior $$p(\theta) \propto \exp\left(-\left(\frac{1}{2\tau_0^2}\right)(\theta-\mu_0)^2\right)$$

$$\exp(a)\exp(b) = \exp(a+b)$$

Posterior $$p(\theta \mid y) \propto \exp\left(-\left(\frac{1}{2}\right)\left[\frac{(y-\theta)^2}{\sigma^2} + \frac{(\theta-\mu_0)^2}{\tau_0^2}\right]\right)$$

Normal distribution – conjugate prior for θ

- Posterior (highly recommended to do BDA3 Ex 2.14a)

$$p(\theta \mid y) \propto \exp\left(-\left(\frac{1}{2}\right)\left[\frac{(y-\theta)^2}{\sigma^2}+\frac{(\theta-\mu_0)^2}{\tau_0^2}\right]\right)$$
$$\propto \exp\left(-\left(\frac{1}{2\tau_1^2}\right)(\theta-\mu_1)^2\right)$$

$$\theta \mid y \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\left(\frac{1}{\tau_0^2}\right)\mu_0 + \left(\frac{1}{\sigma^2}\right)y}{\left(\frac{1}{\tau_0^2}\right)+\left(\frac{1}{\sigma^2}\right)} \quad \text{and} \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}$$

Normal distribution – conjugate prior for θ

- Posterior (highly recommended to do BDA3 Ex 2.14a)

$$p(\theta \mid y) \propto \exp\left(-\left(\frac{1}{2}\right)\left[\frac{(y-\theta)^2}{\sigma^2} + \frac{(\theta-\mu_0)^2}{\tau_0^2}\right]\right)$$
$$\propto \exp\left(-\left(\frac{1}{2\tau_1^2}\right)(\theta-\mu_1)^2\right)$$

$$\theta \mid y \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\left(\frac{1}{\tau_0^2}\right)\mu_0 + \left(\frac{1}{\sigma^2}\right)y}{\left(\frac{1}{\tau_0^2}\right) + \left(\frac{1}{\sigma^2}\right)} \quad \text{and} \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}$$

- 1/variance = precision

Normal distribution – conjugate prior for θ

- Posterior (highly recommended to do BDA3 Ex 2.14a)

$$p(\theta \mid y) \propto \exp\left(-\left(\frac{1}{2}\right)\left[\frac{(y-\theta)^2}{\sigma^2} + \frac{(\theta-\mu_0)^2}{\tau_0^2}\right]\right)$$

$$\propto \exp\left(-\left(\frac{1}{2\tau_1^2}\right)(\theta-\mu_1)^2\right)$$

$$\theta \mid y \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\left(\frac{1}{\tau_0^2}\right)\mu_0 + \left(\frac{1}{\sigma^2}\right)y}{\left(\frac{1}{\tau_0^2}\right) + \left(\frac{1}{\sigma^2}\right)} \quad \text{and} \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}$$

- 1/variance = precision
- Posterior precision = prior precision + data precision

Normal distribution – conjugate prior for θ

- Posterior (highly recommended to do BDA3 Ex 2.14a)

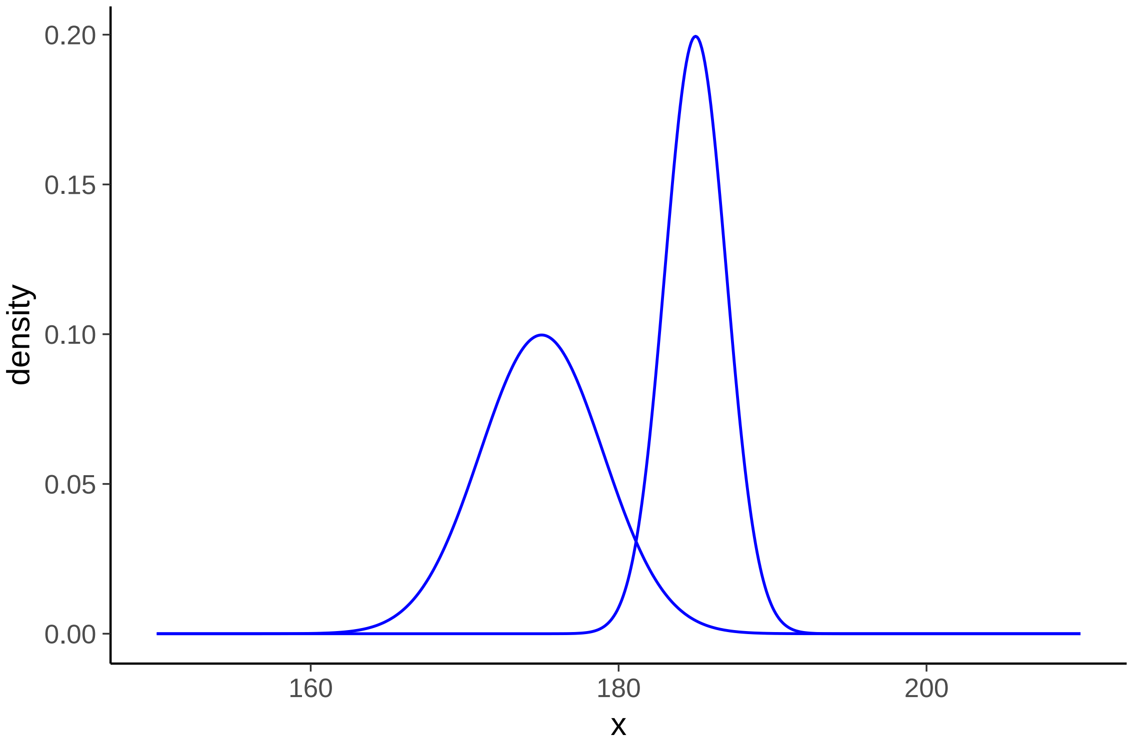
$$p(\theta \mid y) \propto \exp\left(-\left(\frac{1}{2}\right)\left[\frac{(y-\theta)^2}{\sigma^2}+\frac{(\theta-\mu_0)^2}{\tau_0^2}\right]\right)$$

$$\propto \exp\left(-\left(\frac{1}{2\tau_1^2}\right)(\theta-\mu_1)^2\right)$$

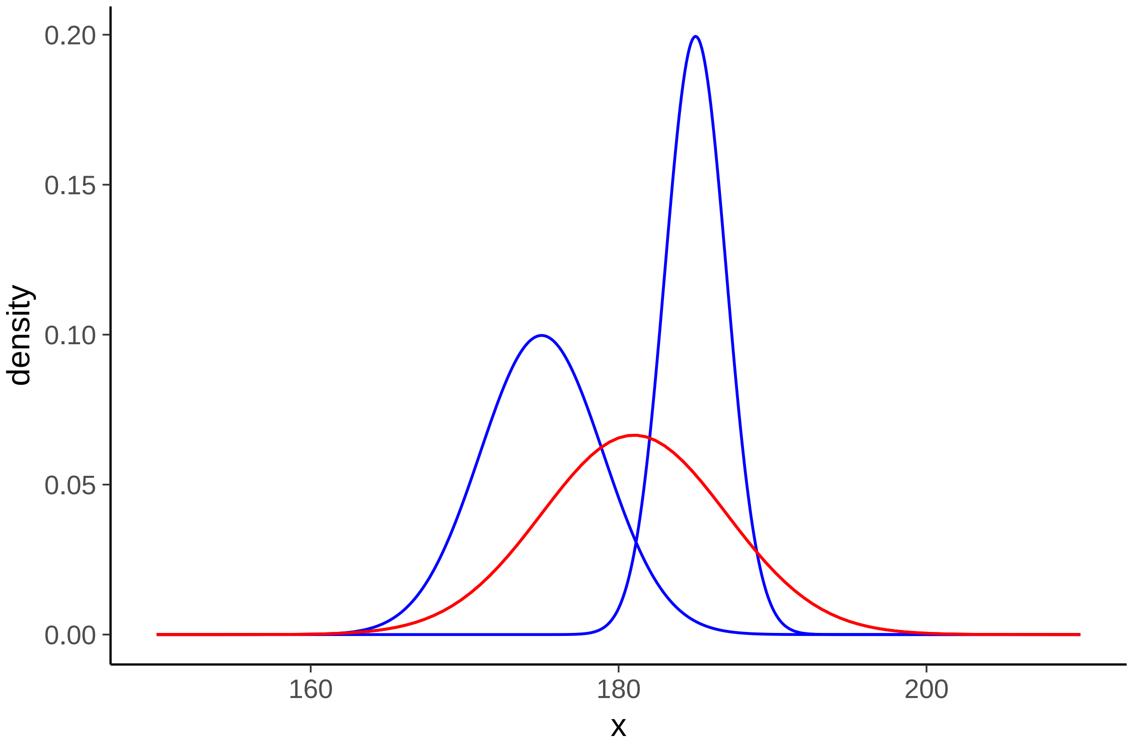
$$\theta \mid y \sim N(\mu_1, \tau_1^2), \quad \text{where} \quad \mu_1 = \frac{\left(\frac{1}{\tau_0^2}\right)\mu_0 + \left(\frac{1}{\sigma^2}\right)y}{\left(\frac{1}{\tau_0^2}\right)+\left(\frac{1}{\sigma^2}\right)} \quad \text{and} \quad \frac{1}{\tau_1^2} = \frac{1}{\tau_0^2}+\frac{1}{\sigma^2}$$

- 1/variance = precision
- Posterior precision = prior precision + data precision
- Posterior mean is precision weighted mean

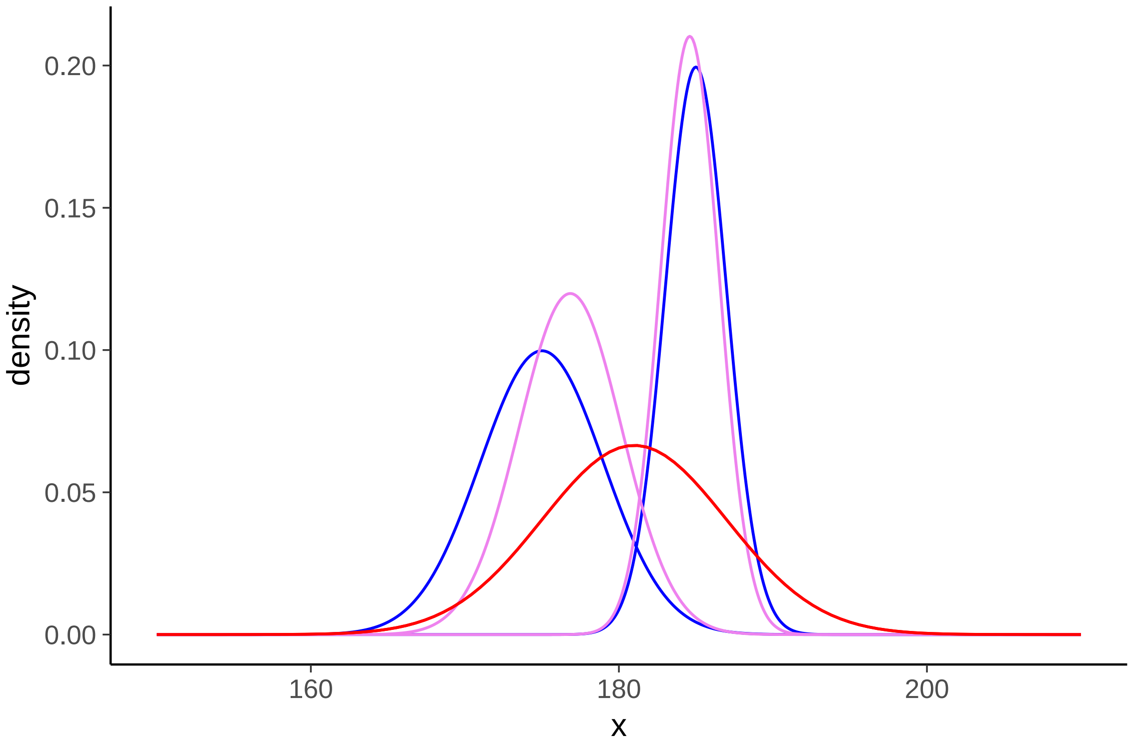
Normal distribution – example



Normal distribution – example



Normal distribution – example



Normal distribution – conjugate prior for θ

- Posterior predictive distribution

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) p(\theta \mid y) \, \mathrm{d}\theta$$

$$p(\tilde{y} \mid y) \propto \int \exp\left(-\left(\frac{1}{2\sigma^2}\right)(\tilde{y} - \theta)^2\right) \exp\left(-\left(\frac{1}{2\tau_1^2}\right)(\theta - \mu_1)^2\right) \mathrm{d}\theta$$

$$\tilde{y} \mid y \sim N(\mu_1, \sigma^2 + \tau_1^2)$$

- Predictive variance = observation model variance σ^2 + posterior variance τ_1^2

Normal model

- Gets more interesting when both mean and variance are unknown
  - next week

Normal model

- Gets more interesting when both mean and variance are unknown
  - next week
- The mean can be also a function of covariates
  - e.g. normal linear regression $y \sim N(\alpha + \beta x, \sigma^2)$

Normal model

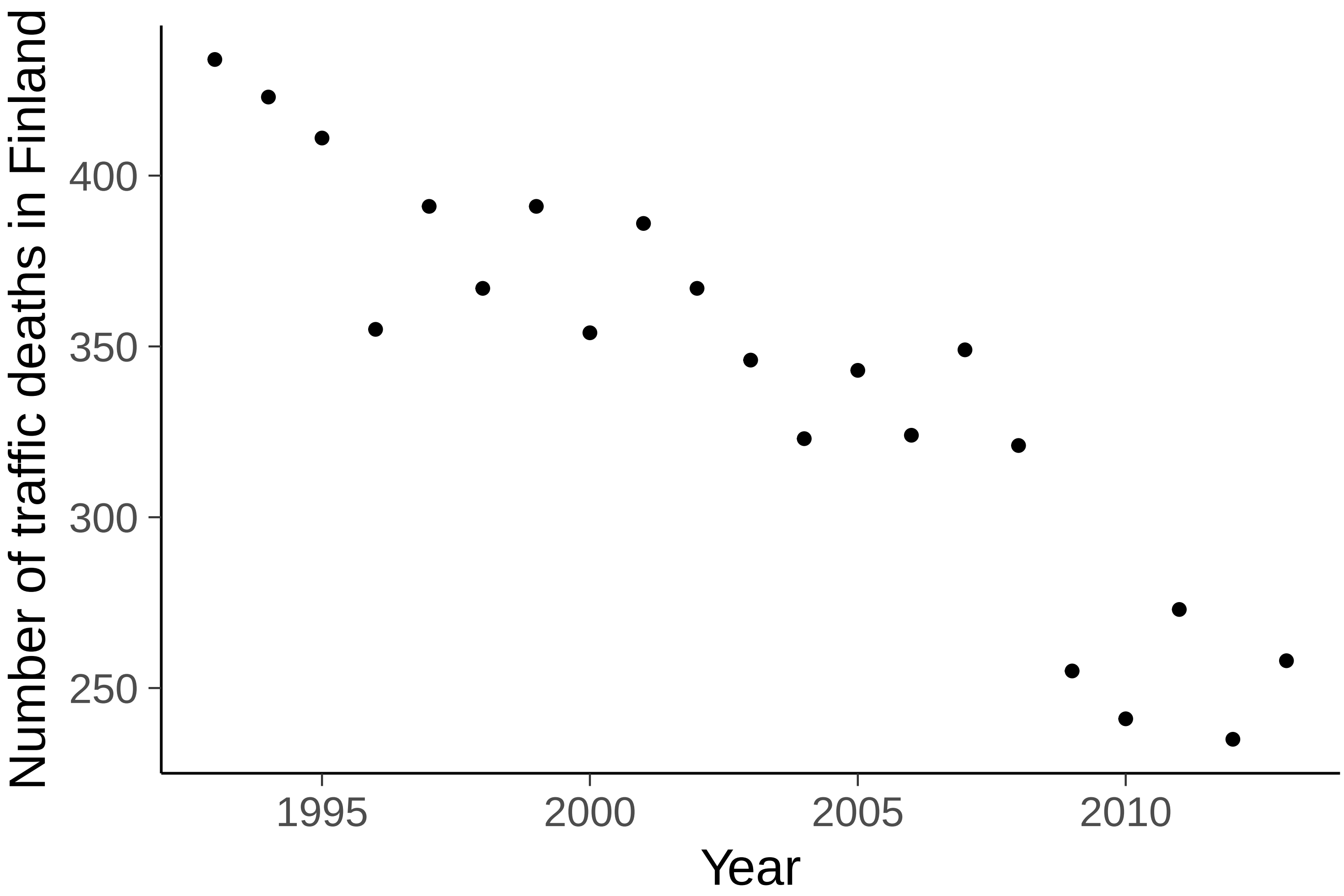
- Gets more interesting when both mean and variance are unknown
  - next week
- The mean can be also a function of covariates
  - e.g. normal linear regression $y \sim N(\alpha + \beta x, \sigma^2)$
- Gaussian processes, Kalman filters, variational inference, Laplace approximation, etc.

Some other one parameter models

- Poisson, useful for count data (e.g. in epidemiology)
- Exponential, useful for time to an event (e.g. particle decay)

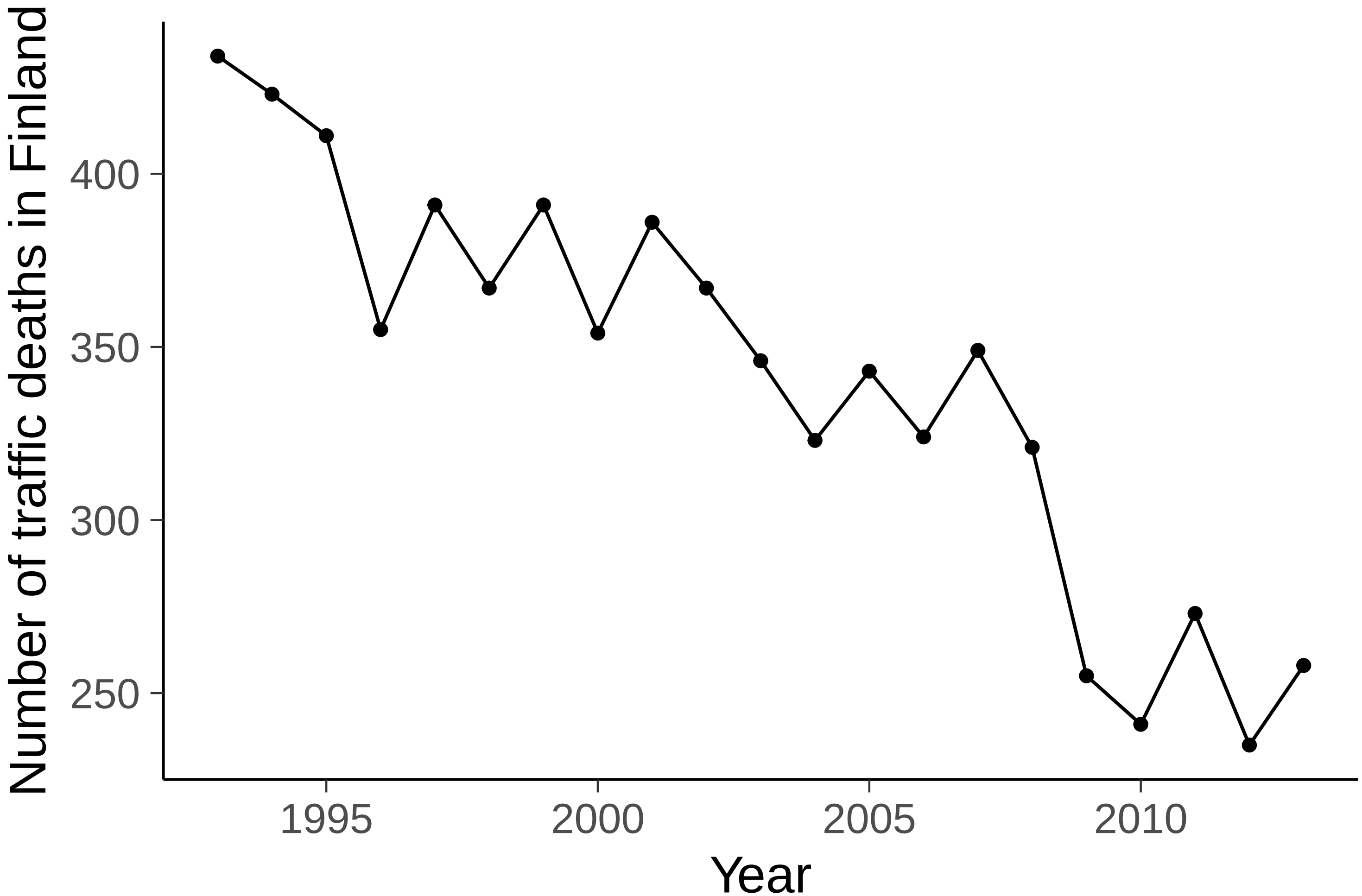
Poisson model for count data

- Number of traffic deaths per year (by Liikenneturva)



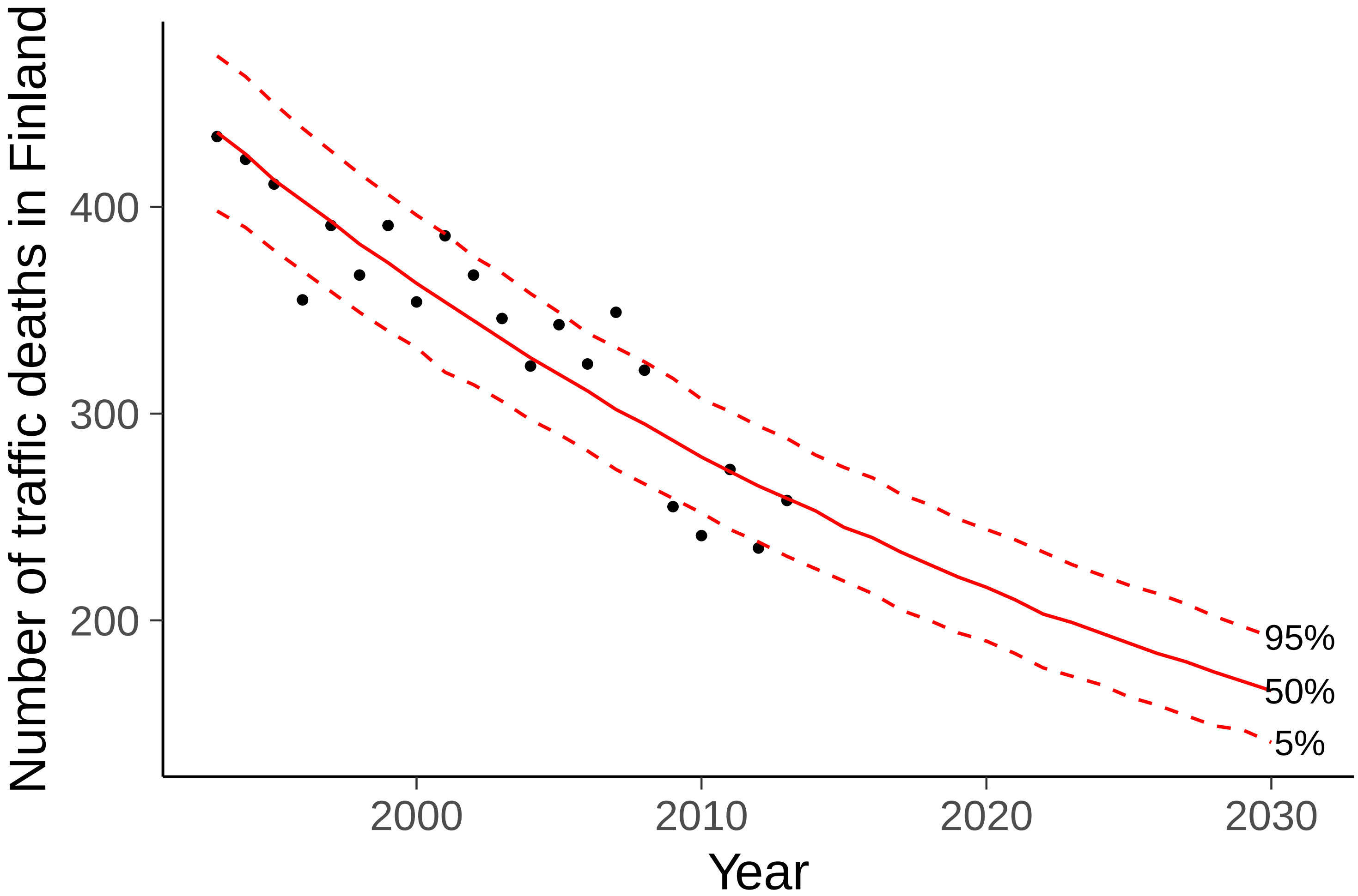
Poisson model for count data

- Number of traffic deaths per year (by Liikenneturva)



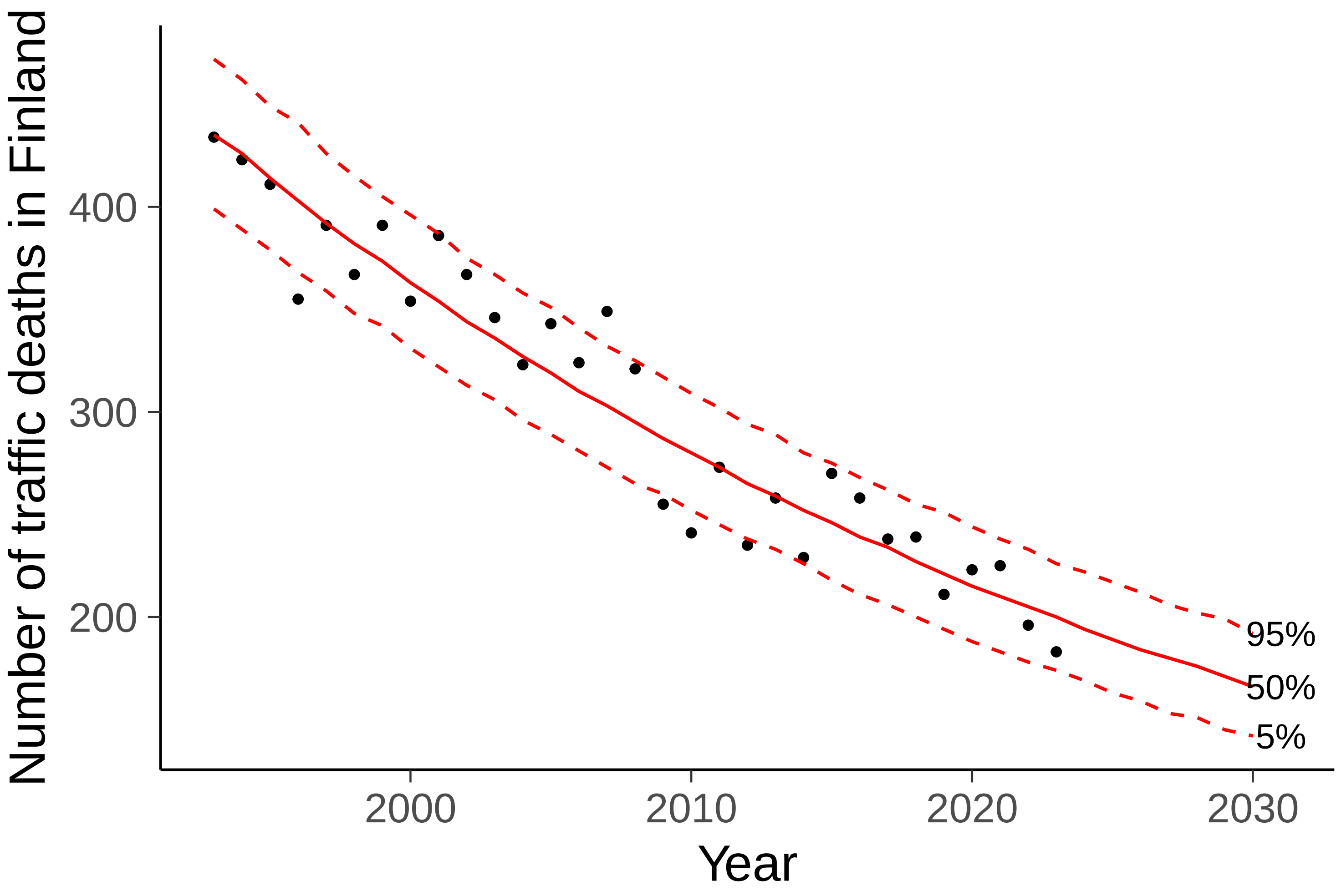
Poisson model for count data

- Number of traffic deaths per year (by Liikenneturva)



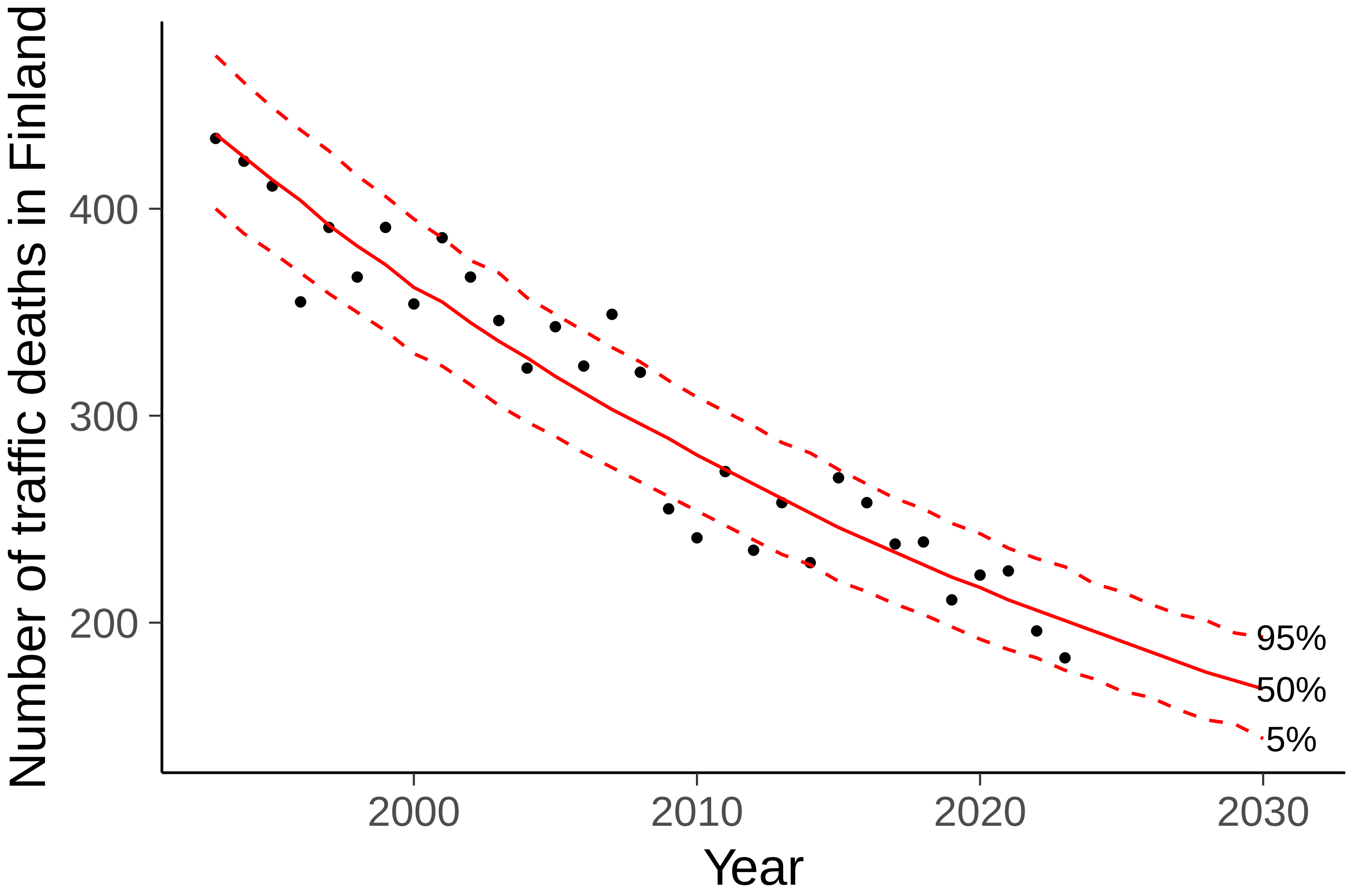
Poisson model for count data

- Number of traffic deaths per year (by Liikenneturva)



Poisson model for count data

- Number of traffic deaths per year (by Liikenneturva)



Thinking priors

- Make a guess of some quantities and then find out useful prior information for that. E.g.
  - proportion of students using MS Windows vs. Apple macOS vs. Linux
  - proportion of students who are taller than 1.9 m
  - proportion of students, who submitted the first assignment, attending the next lecture
  - proportion of students, who submitted the first assignment, submitting the last assignment